

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2023

Analysis Performed by : SL User

Job File : ...\\E-30003-AK.pvdb

Date of Analysis : May 21, 2026 4:49pm

PV Elite 27, January 2025

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2023

Diameter Spec : 1050.000 mm ID	
Vessel Design Length, Tangent to Tangent	728.11 cm
Specified Datum Line Distance	5.08 cm
Shell Side Design Temperature	160 °C
Channel Side Design Temperature	160 °C
Shell Side Design Pressure	85.495 kgf/cm ²
Channel Side Design Pressure	65.765 kgf/cm ²
Shop Shell Side Test Pressure	113.746 kgf/cm ²
Field Shell Side Test Pressure	124.476 kgf/cm ²
Shop Channel Side Test Pressure	90.022 kgf/cm ²
Field Channel Side Test Pressure	100.541 kgf/cm ²

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Normal ized	Impact Tested
Shell	SA-516 70	...	...	K02700	No	Yes
Head	SA-516 60	...	...	K02100	No	Yes
Head	SA-516 70	...	...	K02700	No	Yes
Flange	SA-266 2	...	...	K03506	No	No
Nozzle	SA-106 B	...	...	K03006	No	No
Re-Pad	SA-516 70	...	...	K02700	No	Yes
Nozzle Flg	SA-105	...	...	K03504	No	No
Tubes	SA-179	...	...	K01200	No	No
Tubesheet	SA-266 2	...	...	K03506	No	Yes
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	No	No
Hrz Bolting	SA-193 B7	...	<= 64	G41400	No	No
Saddles	SA-516 70	...	...	K02700	No	No

- Normalized is determined based on the UCS-66 material curve selection and Figure UCS-66.
- Impact Tested is based on material selection and material data properties.

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Dish End	65.870	1.05	70.300	3.1750	No	No
LHS Ch Shell	65.870	1.05	69.200	3.1750	N/A	No
LHS Ch Flg	65.869	1.05	69.200	3.1750	N/A	No
LHS Shell Flg	85.600	1.05	89.100	3.1750	N/A	No
Main Shell	85.600	1.05	89.100	3.1750	N/A	No
RHS Dish End	85.600	1.05	87.500	3.1750	No	No

Liquid Level: 105.00 cm Dens.: 992.645 kg/m³ Sp. Gr.: 0.993

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	5.1	40.0	34.0	32.0	6.1	1.00	1.00
Cylinder	140.0	140.0	30.0	30.0	28.6	7.6	1.00	1.00



Body Flg	173.1	32.5	273.0	265.0	259.9	259.9	1.00	1.00
Body Flg	228.0	33.0	258.0	250.0	246.5	246.5	1.00	1.00
Cylinder	718.0	490.0	38.0	38.0	36.5	10.1	1.00	1.00
Ellipse	723.0	5.1	42.0	36.0	35.2	6.1	1.00	1.00

Local Stress Analysis Results:

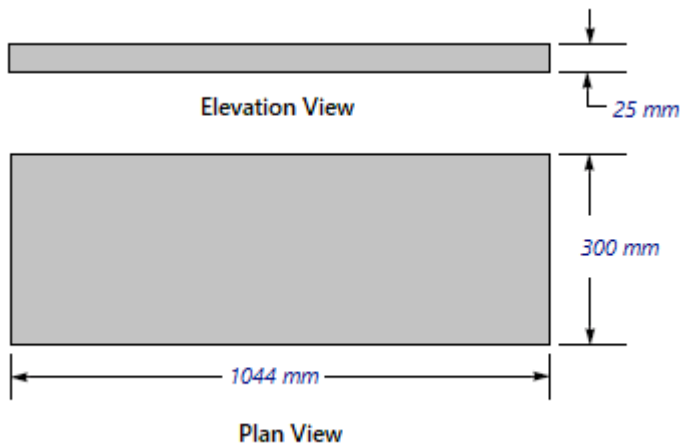
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N-Tube Inlet	WRC-297	0.945	Shell	Passed
N2-Tube Outlet	WRC-297	0.944	Shell	Passed
N3-Shell Inlet	WRC-297	0.954	Shell	Passed
N4-Shell Outle	WRC-297	0.952	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1200.000	mm
Wear Pad Width	425.000	mm
Wear Pad Thickness	16.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	980.000	mm
Baseplate Length	1044.000	mm
Baseplate Thickness	25.000	mm
Baseplate Width	300.000	mm
Number of Ribs (including outside ribs)	4	
Rib Thickness	10.000	mm
Web Thickness	10.000	mm
Height of Center Web	300.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Maximum Tensile Bolt Load 927. kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	38757.37	kgf
Maximum Transverse Saddle Shear Load	1977.84	kgf
Maximum Longitudinal Saddle Shear Load	3955.67	kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	32179.70	kgf
Maximum Transverse Saddle Shear Load	330.24	kgf
Maximum Longitudinal Saddle Shear Load	103.64	kgf

**Weights:**

Fabricated - Bare w/o removable internals	25391.3	kg
Shop Test - Fabricated + water (full)	31138.2	kg
Shipping - Fab. + rem. intls.+ shipping app.	25638.0	kg
Erected - Fab. + rem. intls.+ insul. (etc)	25638.0	kg
Empty - Fab. + intls. + details + wghts.	25638.0	kg
Operating - Empty + operating liquid (no ca)	31347.2	kg
Field Test - Empty weight + test medium (full)	31384.8	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N-Tube Inlet	69.2	OK	...	OK	17.24	OK	Passed
N2-Tube Outlet	69.2	OK	...	OK	17.22	OK	Passed
N3-Shell Inlet	89.1	OK	...	OK	14.15	OK	Passed
N4-Shell Outlet	89.1	OK	...	OK	14.13	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 89.1 Nozzle : N3-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 69.2 Nozzle : N-Tube Inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N-Tube Inlet	750.800	270.000	993.851
20	N2-Tube Outlet	750.800	90.000	993.851
50	N3-Shell Inlet	2830.325	270.000	590.601
50	N4-Shell Outlet	6330.325	90.000	590.601

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3-Shell Inlet	300 mm	80	WNF	323.9	17.450	...	...	398	600
N4-Shell Outlet	300 mm	80	WNF	323.9	17.450	...	...	398	600
N-Tube Inlet	600 mm	160	WNF	609.6	59.512	800.00	48.00	477	600
N2-Tube Outlet	600 mm	160	WNF	609.6	59.512	800.00	48.00	477	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3-Shell In	SA-106 B	38.000	10.000	...	...	...
N4-Shell Ou	SA-106 B	38.000	10.000	...	...	...
N-Tube Inle	SA-106 B	30.000	10.000	36.000	48.000	...
N2-Tube Out	SA-106 B	30.000	10.000	36.000	48.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N3-Shell Inlet	277.953	270.0	300.00	0.00	Main Shell
N4-Shell Outlet	627.953	90.0	300.00	0.00	Main Shell
N-Tube Inlet	70.000	270.0	350.00	0.00	LHS Ch Shell
N2-Tube Outlet	70.000	90.0	350.00	0.00	LHS Ch Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 40.0mm THK X 1050.0mm ID X 5.1cm	SA-516 60
1	WELD NECK FLANGE: 273.0mm THK X 1500.0mm OD	SA-266 2
1	WELD NECK FLANGE: 258.0mm THK X 1500.0mm OD	SA-266 2
1	ELLIPTICAL HEAD: 2.0 X 1, 42.0mm THK X 1050.0mm ID X 5.1cm	SA-516 70
14	BAFFLES: 1042.0mm DIA X 10.0mm THK	SA-516 70
8	TIE RODS: 14mm DIA X 3350mm	SA-193 B7M
1	TUBESHEET: 1500.0mm X 215.0mm THK	SA-266 2
1418	TUBES: 4877.0mm X 19.0mm DIA X 2.1mm THK	SA-179
1	INSULATION: 330mm X 75mm THK	Hot Insulation
2	SADDLE: 200.0mm X 120 DEG	SA-516 70
1	INSULATION: 4900mm X 75mm THK	Hot Insulation
1	INSULATION: 313mm X 75mm THK	Hot Insulation
1	GASKET: 1101.0mm OD X 1082.0mm ID	...
24	BODY FLANGE BOLTS: 80.0mm DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 80.0mm DIA	...
1	CYLINDRICAL SEGMENT 30.0mm THK X 2134.9mm X 140.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 30.0mm THK X 1163.7mm X 140.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 38.0mm THK X 2133.6mm X 490.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 38.0mm THK X 1165.1mm X 490.0cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
LHS Shell Flg[11]	B	10	4		0.896	38.000	1.00	Yes
Main Shell [8]	!	-29	-30	-29	0.980	38.000	1.00	Yes
RHS Dish End [10]	!	-29	-29	-29	0.999	36.000	1.00	Yes
RHS Dish End [7]	!	-29	-36	-29	0.879	42.000	1.00	Yes
N3-Shell Inlet[1]	B	10	8		0.958	38.000	1.00	Yes
Nozzle Flg [5]	A	-18	-28					
N4-Shell Outle[1]	B	10	8		0.956	38.000	1.00	Yes
Nozzle Flg [5]	A	-18	-28					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Tubesheet: SS[13]	!	-29	-35	-29	0.885	53.750	1.00	No
Warmest MDMT:		10	8					
LHS Ch Flg [11]	B	4	-2		0.891	30.000	1.00	No
LHS Dish End [10]	!	-29	-30	-29	0.984	34.000	1.00	Yes
LHS Dish End [7]	!	-29	-37	-29	0.855	40.000	1.00	Yes
LHS Ch Shell [8]	!	-29	-29	-29	0.998	30.000	1.00	No
N-Tube Inlet [1]	B	16	13		0.948	48.000	1.00	Yes
Nozzle Flg [5]	A	-18	-38					
N2-Tube Outlet[1]	B	16	13		0.947	48.000	1.00	Yes
Nozzle Flg [5]	A	-18	-38					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Tubesheet: CS[14]	!	-29	-35	-29	0.885	53.750	1.00	No
Bolting [21]		-40						
Warmest MDMT:		16	13					
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail			
Shell	8.0		-28.9		Fail			
Channel/Tube	13.0		-28.9		Fail			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: LHS Ch Flg and LHS Shell Flg

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	85.493	kgf/cm ²
Channel Side Design Pressure	65.765	kgf/cm ²
Shell Side Design Temperature	160.0	°C
Channel Side Design Temperature	160.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
2.800	0.025
0.000	0.000

Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	350	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM	
Importance Factor for IS-1893	2.000	
Soil Factor	1.000	
Zone Number	2.000	
Percent Seismic for Hydrotest	0.000	
Period of Vibration (User defined) Sec	0.000	



Damping Factor 2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
2.800	0.025
0.000	0.000

Consider Vertical Acceleration Yes
 Design Pressure + Static Head Y
 Consider MAP New and Cold in Noz. Design N
 Consider External Loads for Nozzle Des. Y
 Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element# 1/6
 Element From Node 10
 Element To Node 20
 Element Type Elliptical
 Description LHS Dish End
 Distance "FROM" to "TO" 5.08 cm
 Inside Diameter 1050 mm
 Element Thickness 34 mm
 Internal Corrosion Allowance 3.175 mm
 Nominal Thickness 40 mm



External Corrosion Allowance	0	mm
Design Internal Pressure	65.765	kgf/cm ²
Design Temperature Internal Pressure	160	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 60	[Impact Tested]
Allowable Stress, Ambient	1203.2	kgf/cm ²
Allowable Stress, Operating	1203.2	kgf/cm ²
Allowable Stress, Hydrotest	2018.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	1976.1	kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02100	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	105	cm
Liquid Density	992.63	kg/m ³

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	LHS Ch Shell	
Distance "FROM" to "TO"	140	cm
Inside Diameter	1050	mm
Element Thickness	30	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	30	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	65.765	kgf/cm ²
Design Temperature Internal Pressure	160	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	[Impact Tested]
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2349.3	kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	105	cm
Liquid Density	992.65	kg/m ³
Element From Node	20	



Detail Type	Nozzle	
Detail ID	N-Tube Inlet	
Dist. from "FROM" Node / Offset dist	70	cm
Nozzle Diameter	600	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	826.32	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N2-Tube Outlet	
Dist. from "FROM" Node / Offset dist	70	cm
Nozzle Diameter	600	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	826.32	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	LHS Ch Flg	
Distance "FROM" to "TO"	32.5	cm
Flange Inside Diameter	1050	mm
Element Thickness	265	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	273	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	65.765	kgf/cm ²
Design Temperature Internal Pressure	160	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2294.2	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2220.8	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	105	cm
Liquid Density	992.63	kg/m ³

Element#	4/6	
Element From Node	40	
Element To Node	50	
Element Type	Flange	



Description	LHS Shell Flg	
Distance "FROM" to "TO"	33	cm
Flange Inside Diameter	1050	mm
Element Thickness	250	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	258	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	85.495	kgf/cm ²
Design Temperature Internal Pressure	160	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	105	cm
Liquid Density	999.54	kg/m ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	33	cm
Thickness of Insulation	75	mm
Density	0.00015	kg/cm ³

Element#	5/6	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	490	cm
Inside Diameter	1050	mm
Element Thickness	38	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	38	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	85.495	kgf/cm ²
Design Temperature Internal Pressure	160	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	[Impact Tested]
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2349.3	kgf/cm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	98	cm
Width of Saddle	200	mm
Height of Saddle at Bottom	1200	mm



Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	425	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	50	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	392	cm
Width of Saddle	200	mm
Height of Saddle at Bottom	1200	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	425	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	105	cm
Liquid Density	992.65	kg/m ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Hot Insulation	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	490	cm
Thickness of Insulation	75	mm
Density	0.00015	kg/cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N3-Shell Inlet	
Dist. from "FROM" Node / Offset dist	50	cm
Nozzle Diameter	300	mm
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	270.36	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N4-Shell Outlet	
Dist. from "FROM" Node / Offset dist	400	cm
Nozzle Diameter	300	mm
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	270.36	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



Element#	6/6
Element From Node	60
Element To Node	70
Element Type	Elliptical
Description	RHS Dish End
Distance "FROM" to "TO"	5.08 cm
Inside Diameter	1050 mm
Element Thickness	36 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	42 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	85.495 kgf/cm ²
Design Temperature Internal Pressure	160 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 [Impact Tested]
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	105 cm
Liquid Density	992.65 kg/m ³

Element From Node	60
Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	31.33 cm
Thickness of Insulation	75 mm
Density	0.00015 kg/cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
LHS Dish End		5.08	...	5.08	...
LHS Ch Shell		145.08	...	140	...
LHS Ch Flg		178.215	...	32.5	...
LHS Shell Flg		233.033	...	33	...
Main Shell		723.033	...	490	...
RHS Dish End		728.113	...	5.08	...

PV Elite includes an 1/8 inch (3.175mm) raised face and gasket thicknesses for girth flanges and tubesheet thicknesses where applicable in the Tangent to Tangent length calculation. The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact.

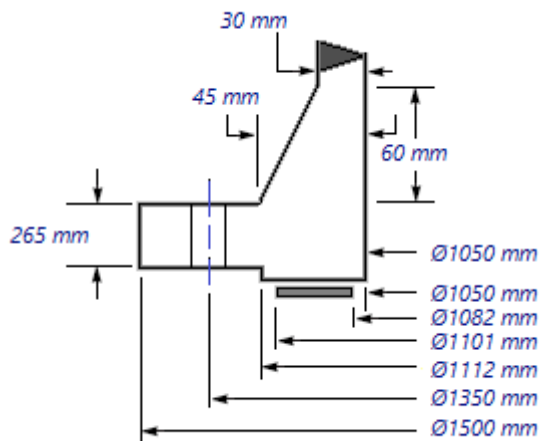
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Flange Input Data Values

Description: LHS Ch Flg :

LHS Ch Flg

Flange Type/Sketch	Integral Weld Neck		
Design Pressure	P	65.87	kgf/cm ²
Design Temperature		160	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1050.000	mm
Flange Outside Diameter	A	1500.000	mm
Flange Thickness	t	265.0000	mm
Thickness of Hub at Small End	go	30.0000	mm
Thickness of Hub at Large End	gl	45.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²
Diameter of Bolt Circle	C	1350.000	mm
Nominal Bolt Diameter	a	80.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1112.000	mm
Flange Face Inside Diameter	Fid	1050.000	mm
Flange Facing Sketch		1, Code Sketch 1a	
Gasket Outside Diameter	Go	1101.000	mm
Gasket Inside Diameter	Gi	1082.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Column for Gasket Seating		2, Code Column II	
Gasket Thickness	tg	3.1750	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (65.9 \cdot (1050/2 + 3.17)) / (1407 \cdot 1 - 0.6 \cdot 65.9) + Ca$$



= 28.6139 mm

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 26.8) / (528 + 0.6 \cdot 26.8)$$

$$= 69.352 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 30) / (525 + 0.6 \cdot 30)$$

$$= 77.742 \text{ kgf/cm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	261.825 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1056.350 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	41.825 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	26.825 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	105.000 mm

Gasket Contact Width, $N = (G_o - G_i) / 2$	9.500 mm
Basic Gasket Width, $b_o = N / 2$	4.750 mm
Effective Gasket Width, $b = b_o$	4.750 mm
Gasket Reaction Diameter, $G = (G_o + G_i) / 2$	1091.500 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi / 4 \cdot G^2 \cdot P_{eq}$$

$$= \pi / 4 \cdot 1092^2 \cdot 65.9$$

$$= 616338.562 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 4.75 \cdot \pi \cdot 1092 \cdot 3 \cdot 65.9$$

$$= 64372.523 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1056^2 \cdot 65.9$$

$$= 577281.438 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 616339 - 577281$$

$$= 39057.105 \text{ kgf}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'p, 0)$$

$$= \max(616339 + 64373 + 0, 0)$$

$$= 680711.062 \text{ kgf}$$

$$= 884613.875 \text{ kgf, Mating Flange Load Governs}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 4.75 \cdot \pi \cdot 1092 + 0 \cdot 0 \cdot 0$$

$$= 114515.766 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max(W_{m1} / S_b, W_{m2} / S_a)$$

$$= \max(884614 / 1621, 114516 / 1621)$$

$$= 545.633 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 80 + 6 \cdot 262 / (3 + 0.5)$$

$$= 608.843 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1350 \cdot \sin(\pi / 24)$$

$$= 176.210 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{176}{2 \cdot 80 + 262}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	545.633	978.440	
Radial Distance between Hub and Bolts:	91.500	105.000	
Radial Distance between Bolts and Edge:	74.000	75.000	
Circ. Spacing between the Bolts:	162.600	176.210	608.843

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 978 \cdot 1621 / (703 \cdot \pi(1101 + 1082))$$

$$= 32.899 \text{ mm [Note: Exceeds actual gasket width, 9.500]}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

$$= 32.000 \text{ mm [Note: Exceeds actual gasket width, 9.500]}$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 978$$

$$= 1586305.88 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wm1 - H$$

$$= 884614 - 616339$$

$$= 268275.34 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1350 - 1092) / 2$$

$$= 129.2500 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (105 + 41.8 + 129) / 2$$

$$= 138.0375 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 105 + (41.8 / 2)$$

$$= 125.9125 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	577281.	125.9125	1.0000	72687.
Face Pressure, Mt	39057.	138.0375	1.0000	5391.
Gasket Load, Mg	268275.	129.2500	1.0000	34675.
Gasket Seating, Matm	1586306.	129.2500	1.0000	205030.

Total Moment for Operation, Mop 112753. kgf-m.
 Total Moment for Gasket seating, Matm 205030. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 168.335 mm
 Hub Ratio, h/h0 = HL / H0 0.356
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.559

Flange Factors for Integral Flange:

Factor F 0.868
 Factor V 0.338



Factor f 1.075

Factors from Figure 2-7.1
 $T = 1.746$
 $Y = 5.704$
 $e = 0.0052 \text{ 1/mm}$
 $K = 1.420$
 $U = 6.268$
 $Z = 2.968$
 $d = 0.22439E+07 \text{ mm}^3$
 Stress Factors
 $\text{Beta} = 2.800$
 $\text{Delta} = 7.999$
 $\text{Alpha} = 2.350$
 $\text{Gamma} = 1.346$
 $\text{Lambda} = 9.345$

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (1.07 \cdot 112753 / 1056) / (9.35 \cdot 41.8^2)$
 $= 701.83 \text{ kgf/cm}^2$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (1.07 \cdot 205030 / 1056) / (9.35 \cdot 41.8^2)$
 $= 1276.22 \text{ kgf/cm}^2$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.8 \cdot 112753 / 1056) / (9.35 \cdot 262^2)$
 $= 46.65 \text{ kgf/cm}^2$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.8 \cdot 205030 / 1056) / (9.35 \cdot 262^2)$
 $= 84.84 \text{ kgf/cm}^2$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (5.7 \cdot 112753 / (262^2 \cdot 1056)) - 2.97 \cdot 46.7$
 $= 749.68 \text{ kgf/cm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (5.7 \cdot 205030 / (262^2 \cdot 1056)) - 2.97 \cdot 84.8$
 $= 1363.22 \text{ kgf/cm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (702 + \max(46.7, 750)) / 2$
 $= 725.76 \text{ kgf/cm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (1276 + \max(84.8, 1363)) / 2$
 $= 1319.72 \text{ kgf/cm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 884614 / 978$
 $= 904.11 \text{ kgf/cm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (114516 / 978)$
 $= 117.04 \text{ kgf/cm}^2$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	701.83	2110.70	1276.22	2110.70
Radial Flange	46.65	1407.13	84.84	1407.13
Tangential Flange	749.68	1407.13	1363.22	1407.13
Maximum Average	725.76	1407.13	1319.72	1407.13
Bolting	904.11	1621.26	117.04	1621.26



Minimum Required Flange Thickness	259.944 mm
Estimated M.A.W.P. (Operating)	69.4 kgf/cm ²
Estimated Finished Weight of Flange at given Thk.	1910.6 kg
Estimated Unfinished Weight of Forging at given Thk	2270.1 kg

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 205030 / 1 \cdot 100000 \cdot 0.34 / (8.88 \cdot 2067025 \cdot 26.8^2 \cdot 168 \cdot 0.3)$
= 0.542 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 112753 / 1 \cdot 100000 \cdot 0.34 / (8.88 \cdot 1984070 \cdot 26.8^2 \cdot 168 \cdot 0.3)$
= 0.310 (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 205030 / 1 \cdot 100000 \cdot 0.34 / (9.35 \cdot 2067025 \cdot 26.8^2 \cdot 168 \cdot 0.3)$
= 0.515 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 112753 / 1 \cdot 100000 \cdot 0.34 / (9.35 \cdot 1984070 \cdot 26.8^2 \cdot 168 \cdot 0.3)$
= 0.295 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.89, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-2 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

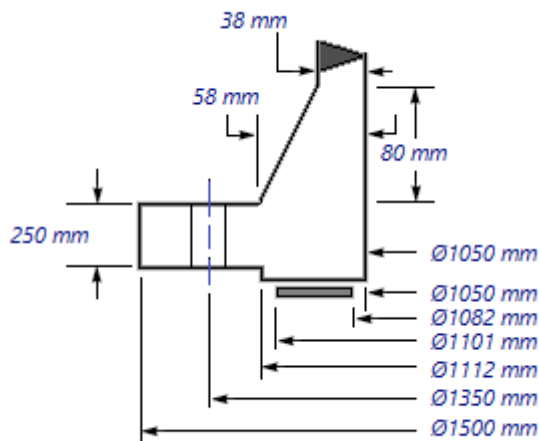
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Flange Input Data Values

Description: LHS Shell Fl :

LHS Shell Flg

Flange Type/Sketch	Integral Weld Neck		
Design Pressure	P	85.60	kgf/cm ²
Design Temperature		160	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1050.000	mm
Flange Outside Diameter	A	1500.000	mm
Flange Thickness	t	250.0000	mm
Thickness of Hub at Small End	go	38.0000	mm
Thickness of Hub at Large End	gl	58.0000	mm
Length of Hub	h	80.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1621.26	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1621.26	kgf/cm ²
Diameter of Bolt Circle	C	1350.000	mm
Nominal Bolt Diameter	a	80.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1112.000	mm
Flange Face Inside Diameter	Fid	1050.000	mm
Flange Facing Sketch		1, Code Sketch 1a	
Gasket Outside Diameter	Go	1101.000	mm
Gasket Inside Diameter	Gi	1082.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	703.07	kgf/cm ²
Column for Gasket Seating		2, Code Column II	
Gasket Thickness	tg	3.1750	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (85.6 \cdot (1050/2 + 3.17)) / (1407 \cdot 1 - 0.6 \cdot 85.6) + Ca$$



$$= 36.5227 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 34.8) / (528 + 0.6 \cdot 34.8)$$

$$= 89.248 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 38) / (525 + 0.6 \cdot 38)$$

$$= 97.610 \text{ kgf/cm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	246.825 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1056.350 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	54.825 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	34.825 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	92.000 mm

Gasket Contact Width, $N = (G_o - G_i) / 2$	9.500 mm
Basic Gasket Width, $b_o = N / 2$	4.750 mm
Effective Gasket Width, $b = b_o$	4.750 mm
Gasket Reaction Diameter, $G = (G_o + G_i) / 2$	1091.500 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi / 4 \cdot G^2 \cdot P_{eq}$$

$$= \pi / 4 \cdot 1092^2 \cdot 85.6$$

$$= 800959.000 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 4.75 \cdot \pi \cdot 1092 \cdot 3 \cdot 85.6$$

$$= 83654.906 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 1056^2 \cdot 85.6$$

$$= 750202.562 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 800959 - 750203$$

$$= 50756.434 \text{ kgf}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'p, 0)$$

$$= \max(800959 + 83655 + 0, 0)$$

$$= 884613.875 \text{ kgf}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 703 \cdot 4.75 \cdot \pi \cdot 1092 + 0 \cdot 0 \cdot 0$$

$$= 114515.766 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max(W_{m1} / S_b, W_{m2} / S_a)$$

$$= \max(884614 / 1621, 114516 / 1621)$$

$$= 545.633 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 80 + 6 \cdot 247 / (3 + 0.5)$$

$$= 583.129 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1350 \cdot \sin(\pi / 24)$$

$$= 176.210 \text{ mm}$$



ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{176}{(2 \cdot 80 + 247)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	545.633	978.440	
Radial Distance between Hub and Bolts:	91.500	92.000	
Radial Distance between Bolts and Edge:	74.000	75.000	
Circ. Spacing between the Bolts:	162.600	176.210	583.129

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 978 \cdot 1621 / (703 \cdot \pi(1101 + 1082))$$

$$= 32.899 \text{ mm [Note: Exceeds actual gasket width, 9.500]}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

$$= 32.000 \text{ mm [Note: Exceeds actual gasket width, 9.500]}$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= Sa \cdot Ab$$

$$= 1621 \cdot 978$$

$$= 1586305.88 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wm1 - H$$

$$= 884614 - 800959$$

$$= 83654.90 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1350 - 1092) / 2$$

$$= 129.2500 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (92 + 54.8 + 129) / 2$$

$$= 138.0375 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 92 + (54.8 / 2)$$

$$= 119.4125 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	750203.	119.4125	1.0000	89584.
Face Pressure, Mt	50756.	138.0375	1.0000	7006.
Gasket Load, Mg	83655.	129.2500	1.0000	10812.
Gasket Seating, Matm	1586306.	129.2500	1.0000	205030.

Total Moment for Operation, Mop 107402. kgf-m.
 Total Moment for Gasket seating, Matm 205030. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 191.800 mm
 Hub Ratio, h/h0 = HL / H0 0.417
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.574

Flange Factors for Integral Flange:

Factor F 0.857
 Factor V 0.316
 Factor f 1.000



Factors from Figure 2-7.1
 $T = 1.746$
 $Y = 5.704$
 $e = 0.0045 \text{ 1/mm}$
 $K = 1.420$
 $U = 6.268$
 $Z = 2.968$
 $d = 0.46084E+07 \text{ mm}^3$
 Stress Factors
 $Beta = 2.471$
 $Delta = 3.263$
 $Alpha = 2.103$
 $Gamma = 1.205$
 $Lambda = 4.468$

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g1^2)$
 $= (1 \cdot 107402 / 1056) / (4.47 \cdot 54.8^2)$
 $= 757.09 \text{ kgf/cm}^2$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g1^2)$
 $= (1 \cdot 205030 / 1056) / (4.47 \cdot 54.8^2)$
 $= 1445.28 \text{ kgf/cm}^2$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.47 \cdot 107402 / 1056) / (4.47 \cdot 247^2)$
 $= 92.30 \text{ kgf/cm}^2$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.47 \cdot 205030 / 1056) / (4.47 \cdot 247^2)$
 $= 176.19 \text{ kgf/cm}^2$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (5.7 \cdot 107402 / (247^2 \cdot 1056)) - 2.97 \cdot 92.3$
 $= 678.03 \text{ kgf/cm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (5.7 \cdot 205030 / (247^2 \cdot 1056)) - 2.97 \cdot 176$
 $= 1294.35 \text{ kgf/cm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (757 + \max(92.3, 678)) / 2$
 $= 717.56 \text{ kgf/cm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (1445 + \max(176, 1294)) / 2$
 $= 1369.81 \text{ kgf/cm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 884614 / 978$
 $= 904.11 \text{ kgf/cm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (114516 / 978)$
 $= 117.04 \text{ kgf/cm}^2$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	757.09	2110.70	1445.28	2110.70
Radial Flange	92.30	1407.13	176.19	1407.13
Tangential Flange	678.03	1407.13	1294.35	1407.13
Maximum Average	717.56	1407.13	1369.81	1407.13
Bolting	904.11	1621.26	117.04	1621.26



Minimum Required Flange Thickness	246.456 mm
Estimated M.A.W.P. (Operating)	89.2 kgf/cm ²
Estimated Finished Weight of Flange at given Thk.	1848.9 kg
Estimated Unfinished Weight of Forging at given Thk	2305.0 kg

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 205030 / 1 \cdot 100000 \cdot 0.32 / (4.32 \cdot 2067025 \cdot 34.8^2 \cdot 192 \cdot 0.3)$
= 0.542 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 107402 / 1 \cdot 100000 \cdot 0.32 / (4.32 \cdot 1984070 \cdot 34.8^2 \cdot 192 \cdot 0.3)$
= 0.296 (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 205030 / 1 \cdot 100000 \cdot 0.32 / (4.47 \cdot 2067025 \cdot 34.8^2 \cdot 192 \cdot 0.3)$
= 0.524 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 107402 / 1 \cdot 100000 \cdot 0.32 / (4.47 \cdot 1984070 \cdot 34.8^2 \cdot 192 \cdot 0.3)$
= 0.286 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.9, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	10 °C
Min Metal Temp. at Required thickness (UCS 66.1)	4 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
LHS Dish End		65.87	40	3.175	1050	1203.2	1
LHS Ch Shell		65.87	30	3.175	1050	1407.1	2
LHS Ch Flg		65.869	273	3.175	1050	1407.1	3
LHS Shell Flg		85.6	258	3.175	1050	1407.1	4
Main Shell		85.6	38	3.175	1050	1407.1	5
RHS Dish End		85.6	42	3.175	1050	1407.1	6

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
LHS Dish End		65.765	70.3	77.4	34	32.0182	1
LHS Ch Shell		65.765	69.2	77.7	30	28.6141	2
LHS Ch Flg		65.765	69.2	77.7	265	259.944	3
LHS Shell Flg		85.495	89.1	97.6	250	246.456	4
Main Shell		85.495	89.1	97.6	38	36.5225	5
RHS Dish End		85.495	87.5	95.8	36	35.2452	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

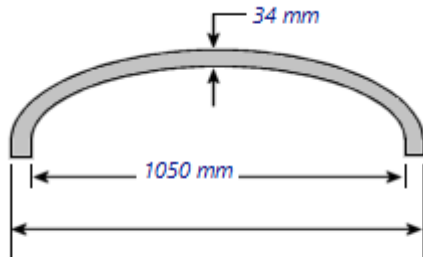
Shell Side MAWP = 87.5 kgf/cm²
 Shell Side MAPnc = 95.8 kgf/cm²
 Channel Side MAWP = 69.2 kgf/cm²
 Channel Side MAPnc = 77.4 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Dish End

Elliptical Head Design Information:							
Design Pressure	P	65.870	kgf/cm²	Int. Design Temperature		160	°C
Material		SA-516	60	Ext. Design Temperature		160	°C
Length		5.080	cm	External Pressure Chart		CS-2	
				UNS Number		K02100	
Int. Corr. All.	c	3.1750	mm	Allowable Stress (ope)	S	1203.2	kgf/cm²
Ext. Corr. All.	ce	0.0000	mm	Allowable Stress (amb)	Sa	1203.2	kgf/cm²
Inside Diameter	Di	1050.000	mm	Actual Stress		1126.3	kgf/cm²
Spec. Min. Thk t,e, or sa		34.0000	mm	Spec. Nominal Thk.		40.000	mm
Surface Area		15644.8	cm²	SG of Contents		0.993	
Element Volume		195520.6	cm³	Weight of Contents		194.1	kgf
Empty Weight		494.8	kgf	Operating Weight		688.8	kgf
				Comp. All. Ope. Stress		1443.8	kgf/cm²
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (65.9 \cdot 1056 \cdot 0.99) / (2 \cdot 1203 \cdot 1 - 0.2 \cdot 65.9)$$

$$= 28.8432 + 3.1750 = 32.0182 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.105 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1203 \cdot 1 \cdot 30.8) / (0.99 \cdot 1056 + 0.2 \cdot 30.8)$$

$$= 70.4 - 0.1 = 70.3 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1203 \cdot 1 \cdot 34) / (1 \cdot 1050 + 0.2 \cdot 34)$$

$$= 77.4 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (65.9 \cdot (0.99 \cdot 1056 + 0.2 \cdot 30.8)) / (2 \cdot 1 \cdot 30.8)$$

$$= 1126.272 \text{ kgf/cm}^2$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (65.9 \cdot 528) / (1203 \cdot 1 - 0.6 \cdot 65.9) + 3.17$
 $= 33.072 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
Less Operating Hydrostatic Head Pressure of 0.105 kgf/cm²
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1203 \cdot 1 \cdot 36.8) / (528 + 0.6 \cdot 36.8)$
 $= 80.5 - 0.1 = 80.4 \text{ kgf/cm}^2$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1056 / (2 \cdot 266))^2) / 6$
 $= 0.992057$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 15.113 %
Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

EXTERNAL PRESSURE RESULTS for: LHS Dish End

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
30.825	1118.00	36.27	0.0038294	1153.68

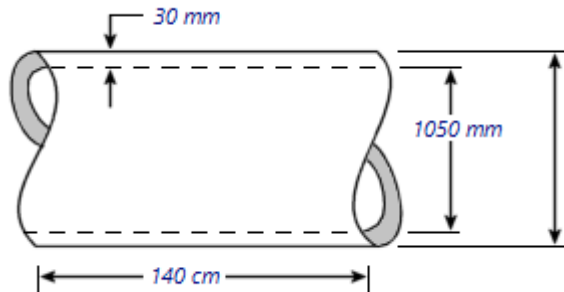
$$MAEP = B / (K_0 \cdot Do / t) = 1154 / (0.9 \cdot 36.3) = 35.3 \text{ kgf/cm}^2$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.905	1118.00	384.89	0.0003609	365.34

$$MAEP = B / (K_0 \cdot Do / t) = 365 / (0.9 \cdot 385) = 1.05 \text{ kgf/cm}^2$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	65.870 kgf/cm²	Int. Design Temperature	160 °C	
Material		SA-516 70	Ext. Design Temperature	160 °C	
Length		140.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1050.000 mm	Actual Stress	1336.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		30.0000 mm	Spec. Nominal Thk.	30.000 mm	
Surface Area		48820.4 cm²	SG of Contents	0.993	
Element Volume		1212262.2 cm³	Weight of Contents	1203.4 kgf	
Empty Weight		1104.4 kgf	Operating Weight	2307.8 kgf	
			Comp. All. Ope. Stress	1453.1 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (65.9 \cdot 528) / (1407 \cdot 1 - 0.6 \cdot 65.9)$$

$$= 25.4391 + 3.1750 = 28.6141 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.105 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 26.8) / (528 + 0.6 \cdot 26.8)$$

$$= 69.4 - 0.1 = 69.2 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 30) / (525 + 0.6 \cdot 30)$$

$$= 77.7 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (65.9 \cdot (528 + 0.6 \cdot 26.8)) / (1 \cdot 26.8)$$

$$= 1336.470 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.778 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
26.825	1110.00	1863.30	41.38	1.6786	0.0029182	1108.74

$$MAEP = (4*B)/(3*(Do/t)) = (4*1109)/(3*41.4) = 35.7 \text{ kgf/cm}^2$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.438	1110.00	1863.30	250.12	1.6786	0.0001954	197.85

$$MAEP = (4*B)/(3*(Do/t)) = (4*198)/(3*250) = 1.05 \text{ kgf/cm}^2$$

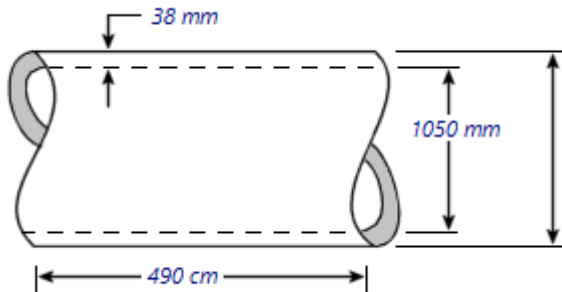
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
26.825	1110.00	49387.31	41.38	44.4931	0.0006483	656.37

$$MAEP = (4*B)/(3*(Do/t)) = (4*656)/(3*41.4) = 21.1 \text{ kgf/cm}^2$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	85.600 kgf/cm²	Int. Design Temperature	160 °C	
Material		SA-516 70	Ext. Design Temperature	160 °C	
Length		490.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	1050.000 mm	Actual Stress	1349.6 kgf/cm²	
Spec. Min. Thk t,e, or sa		38.0000 mm	Spec. Nominal Thk.	38.000 mm	
Surface Area		173334.2 cm²	SG of Contents	0.993	
Element Volume		2169467.0 cm³	Weight of Contents	2153.5 kgf	
Empty Weight		4932.7 kgf	Operating Weight	7086.2 kgf	
			Comp. All. Ope. Stress	1478.9 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (85.6 \cdot 528) / (1407 \cdot 1 - 0.6 \cdot 85.6)$$

$$= 33.3475 + 3.1750 = 36.5225 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.105 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 34.8) / (528 + 0.6 \cdot 34.8)$$

$$= 89.2 - 0.1 = 89.1 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407 \cdot 1 \cdot 38) / (525 + 0.6 \cdot 38)$$

$$= 97.6 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (85.6 \cdot (528 + 0.6 \cdot 34.8)) / (1 \cdot 34.8)$$

$$= 1349.609 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 3.493 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
34.825	1126.00	5368.30	32.33	4.7676	0.0013700	941.11

$$MAEP = (4*B)/(3*(Do/t)) = (4*941)/(3*32.3) = 38.8 \text{ kgf/cm}^2$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.973	1126.00	5368.30	161.49	4.7676	0.0001262	127.74

$$MAEP = (4*B)/(3*(Do/t)) = (4*128)/(3*161) = 1.05 \text{ kgf/cm}^2$$

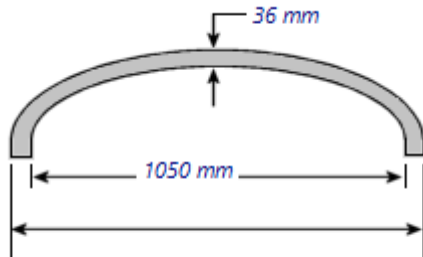
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
34.825	1126.00	190375.48	32.33	50.0000	0.0010734	873.68

$$MAEP = (4*B)/(3*(Do/t)) = (4*874)/(3*32.3) = 36 \text{ kgf/cm}^2$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Dish End

Elliptical Head Design Information:							
Design Pressure		P	85.600 kgf/cm ²	Int. Design Temperature		160 °C	
Material			SA-516 70	Ext. Design Temperature		160 °C	
Length			5.080 cm	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.		c	3.1750 mm	Allowable Stress (ope)		S	1407.1 kgf/cm ²
Ext. Corr. All.		ce	0.0000 mm	Allowable Stress (amb)		Sa	1407.1 kgf/cm ²
Inside Diameter		Di	1050.000 mm	Actual Stress		1375.0 kgf/cm ²	
Spec. Min. Thk t,e, or sa			36.0000 mm	Spec. Nominal Thk.		42.000 mm	
Surface Area			15749.4 cm ²	SG of Contents		0.993	
Element Volume			195520.6 cm ³	Weight of Contents		194.1 kgf	
Empty Weight			521.3 kgf	Operating Weight		715.4 kgf	
				Comp. All. Ope. Stress		1478.9 kgf/cm ²	
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (85.6 \cdot 1056 \cdot 0.99) / (2 \cdot 1407 \cdot 1 - 0.2 \cdot 85.6)$$

$$= 32.0702 + 3.1750 = 35.2452 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.105 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407 \cdot 1 \cdot 32.8) / (0.99 \cdot 1056 + 0.2 \cdot 32.8)$$

$$= 87.6 - 0.1 = 87.5 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407 \cdot 1 \cdot 36) / (1 \cdot 1050 + 0.2 \cdot 36)$$

$$= 95.8 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (85.6 \cdot (0.99 \cdot 1056 + 0.2 \cdot 32.8)) / (2 \cdot 1 \cdot 32.8)$$

$$= 1374.970 \text{ kgf/cm}^2$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (85.6 \cdot 528) / (1407 \cdot 1 - 0.6 \cdot 85.6) + 3.17$
 $= 36.523 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
[Less Operating Hydrostatic Head Pressure of 0.105 kgf/cm²](#)
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (1407 \cdot 1 \cdot 38.8) / (528 + 0.6 \cdot 38.8)$
 $= 99.1 - 0.1 = 99 \text{ kgf/cm}^2$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$
 $= (2 + (1056 / (2 \cdot 266)))^2 / 6$
 $= 0.992057$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 15.789 %
[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

[Note: This Element/Detail was specified as being Impact Tested.](#)

MDMT Calculations in the Head Straight Flange:

[Note: This Element/Detail was specified as being Impact Tested.](#)

[Note:](#)
[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.125 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.563 mm

EXTERNAL PRESSURE RESULTS for: RHS Dish End

Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 2024859 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
32.825	1122.00	34.18	0.0040633	1162.42

$$\text{MAEP} = B / (K_0 \cdot Do / t) = 1162 / (0.9 \cdot 34.2) = 37.8 \text{ kgf/cm}^2$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.915	1122.00	384.89	0.0003609	365.34

$$\text{MAEP} = B / (K_0 \cdot Do / t) = 365 / (0.9 \cdot 385) = 1.05 \text{ kgf/cm}^2$$

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	113.746 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	111.143 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	124.476 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	96.247 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	122.258 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	122.258 kgf/cm ²
App 27-4	= M.A.W.P.	87.497 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	90.022 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	85.494 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	100.541 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	76.172 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	94.044 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	94.044 kgf/cm ²
App 27-4	= M.A.W.P.	69.247 kgf/cm ²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	1400.7	2018.9	0.694	90.13
LHS Ch Shell	1631.3	2386.0	0.684	90.13
Main Shell	1641.3	2386.0	0.688	113.85
RHS Dish End	1671.7	2386.0	0.701	113.85

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N-Tube Inlet	Nozzle	1203.20	1203.20	1.000
N-Tube Inlet	Pad	1407.13	1407.13	1.000
N2-Tube Outlet	Nozzle	1203.20	1203.20	1.000
N2-Tube Outlet	Pad	1407.13	1407.13	1.000
N3-Shell Inlet	Nozzle	1203.20	1203.20	1.000
N4-Shell Outlet	Nozzle	1203.20	1203.20	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
LHS Dish End	1203.20	1203.20	1.000
LHS Ch Shell	1407.13	1407.13	1.000
LHS Ch Flg	1407.13	1407.13	1.000
LHS Shell Flg	1407.13	1407.13	1.000
Main Shell	1407.13	1407.13	1.000
RHS Dish End	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	942.17	942.17	1.000



Tubesheet Material	1407.13	1407.13	1.000
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Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N-Tube Inlet	425.55	1564.16	0.272
N2-Tube Outlet	425.55	1564.16	0.272
N3-Shell Inlet	1010.94	1564.16	0.646
N4-Shell Outlet	1010.94	1564.16	0.646

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	113.746 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	111.143 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	124.476 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	96.247 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	122.258 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	122.258 kgf/cm ²
App 27-4	= M.A.W.P.	87.497 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	90.022 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	85.494 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	100.541 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	76.172 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	94.044 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	94.044 kgf/cm ²
App 27-4	= M.A.W.P.	69.247 kgf/cm ²

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	1720.9	2018.9	0.852	100.65
LHS Ch Shell	2042.1	2386.0	0.856	100.65
Main Shell	1964.2	2386.0	0.823	124.58
RHS Dish End	2001.1	2386.0	0.839	124.58

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N-Tube Inlet	Nozzle	1203.20	1203.20	1.000
N-Tube Inlet	Pad	1407.13	1407.13	1.000
N2-Tube Outlet	Nozzle	1203.20	1203.20	1.000
N2-Tube Outlet	Pad	1407.13	1407.13	1.000
N3-Shell Inlet	Nozzle	1203.20	1203.20	1.000
N4-Shell Outlet	Nozzle	1203.20	1203.20	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
LHS Dish End	1203.20	1203.20	1.000
LHS Ch Shell	1407.13	1407.13	1.000
LHS Ch Flg	1407.13	1407.13	1.000
LHS Shell Flg	1407.13	1407.13	1.000
Main Shell	1407.13	1407.13	1.000
RHS Dish End	1407.13	1407.13	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	942.17	942.17	1.000



Tubesheet Material	1407.13	1407.13	1.000
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Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
N-Tube Inlet	475.22	1564.16	0.304
N2-Tube Outlet	475.22	1564.16	0.304
N3-Shell Inlet	1106.22	1564.16	0.707
N4-Shell Outlet	1106.22	1564.16	0.707

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	1118	30.825	0.0038294	1153.68
20	30	186.33	1110	26.825	0.0029182	1108.74
30	40	No Calc	...	261.825	No Calc	No Calc
40	50	No Calc	...	246.825	No Calc	No Calc
50	60	536.83	1126	34.825	0.00137	941.108
60	70	No Calc	1122	32.825	0.0040633	1162.42

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	Design External Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	34	6.07974	1.05461	35.3
20	30	30	7.61284	1.05461	35.7
30	40	265	259.944	1.05461	No Calc
40	50	250	246.456	1.05461	No Calc
50	60	38	10.1477	1.05461	38.8
60	70	36	6.09014	1.05461	37.8

Minimum

35.3

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	186.33	4938.73	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	536.83	19037.5	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	494.764	195521	455.492	198820	49.4764
20	30	1104.45	1212262	990.464	1226969	110.445
30	40	1966.48	291213	1939.36	291843	196.648
40	50	1904.78	295385	1877.27	296225	190.478
50	60	4932.66	2169467	4533.71	2220941	493.266
60	70	521.306	195521	481.898	198820	52.1306
Total		10924	4359368.50	10278	4433619.50	1092



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	194.082	-8.75	0.36335E-05	...	Liquid: 10
20	Liqd	1203.36	70	...	...	Liquid: 20
20	Nozl	908.947	70	-77.0288	...	N-Tube Inlet
20	Nozl	908.947	70	77.0288	...	N2-Tube Outlet
30	Liqd	289.071	16.25	...	...	Liquid: 30
40	Liqd	295.253	16.5	...	...	Liquid: 40
40	Insl	15.2244	16.5	...	...	Hot Insulation
50	Sadl	265.977	98	83.55	...	Left Saddle
50	Sadl	265.977	392	83.55	...	New Saddle
50	Liqd	2153.53	245	...	...	Liquid: 50
50	Insl	208.229	245	...	...	Hot Insulation
50	Nozl	297.391	50	-66.9475	...	N3-Shell Inlet
50	Nozl	297.391	400	66.9475	...	N4-Shell Outlet
60	Liqd	194.084	13.83	0.36335E-05	...	Liquid: 60
60	Insl	23.1945	15.665	...	...	Hot Insulation
30	FTsh	2498.3	43.885	...	...	
30	Tube	7454.15	276.985	...	...	Tubes
40	Bafl	34.0682	76.2	...	...	Baffles
40	Bafl	34.0682	107.6	...	...	Baffles
40	Bafl	34.0682	139	...	...	Baffles
40	Bafl	34.0682	170.4	...	...	Baffles
40	Bafl	34.0682	201.8	...	...	Baffles
40	Bafl	34.0682	233.2	...	...	Baffles
40	Bafl	34.0682	264.6	...	...	Baffles
40	Bafl	34.0682	296	...	...	Baffles
40	Bafl	34.0682	327.4	...	...	Baffles
40	Bafl	34.0682	358.8	...	...	Baffles
40	Bafl	34.0682	390.2	...	...	Baffles
40	Bafl	34.0682	421.6	...	...	Baffles
40	Bafl	34.0682	453	...	...	Baffles
40	Bafl	34.4928	484.4	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	532.0
Liquid	4329.4
Insulation	246.6
Nozzles	2412.7
Exchanger Components	10429.8
Liquid in Tubes	1379.8
Sum of the Detail Weights	19330.3 kg



Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	12016.9	12016.9	12016.9	12016.9	12016.9	12016.9
Saddles	532.0	532.0	532.0	532.0	532.0	532.0
Nozzles	2412.7	2412.7	2412.7	2412.7	2412.7	2412.7
Exchanger	10429.8	10429.8	10429.8	10429.8	10429.8	10429.8
Insulation	...	...	246.6	246.6	246.6	246.6
Ope. Liquid	...	...	...	...	...	4329.4
Tube Ope Lqd	...	...	...	...	...	1379.8
Test Liquid	...	4357.4	...	...	...	...
Tube Tst Lqd	...	1389.4	...	...	...	...
Totals	25391.3	31138.2	25638.0	25638.0	25638.0	31347.2

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	25391.3 kg
Shop Test Wt.	- Fabricated weight + water (full)	31138.2 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	25638.0 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	25638.0 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	25638.0 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	31347.2 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	30636.3 kg
Field Test Wt.	- Empty weight + test medium (full)	31384.8 kg

Exchanger Tube Data

Volume of Exchanger tubes :	1390057.5 cm ³
Weight of Ope Liq in tubes :	1379.8 kg
Weight of Water in tubes :	1389.4 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	15644.8	13626.7
20	30	48820.4	46181.4
30	40	24026.1	10984.6
40	50	24100.9	11149.5
50	60	173334	161635
60	70	15749.4	13626.7

Total	301675.8	257203.8 cm ²
Total	324.7	276.9 ft ²



Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N-Tube Inlet	91.45	104.12	160	600	GR 1.1	16.459	91.450	91.450	91.450
N2-Tube Outlet	91.45	104.12	160	600	GR 1.1	16.459	91.450	91.450	91.450
N3-Shell Inlet	91.45	104.12	160	600	GR 1.1	27.035	91.450	64.415	91.450
N4-Shell Outle	91.45	104.12	160	600	GR 1.1	27.035	91.450	64.415	91.450

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 91.450 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 104.115 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 91.450 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 104.115 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
N-Tube Inlet	657.225	0.50	9174	0	69.143	91.450	Pass
N2-Tube Outlet	657.225	0.50	9174	0	69.247	91.450	Pass
N3-Shell Inlet	350.901	0.50	2294	0	87.393	91.450	Pass
N4-Shell Outlet	350.901	0.50	2294	0	87.497	91.450	Pass

Evaluating Flange on Nozzle [N-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(9174) + 4 \cdot \text{abs}(0)657.225 \leq \pi \cdot 657.225^3 \cdot [(91.450 - 69.143) + .500 \cdot 91.450]$$

$$= 0.14679 + 06 \leq 0.60675 + 06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(91.450, 0.500 \cdot 91.450 + 91.450 - 146787.0 / (\pi \cdot 657.225^3))$$

$$= \min(91.450, 120.717)$$

$$= 91.450 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N2-Tube Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(9174) + 4 \cdot \text{abs}(0)657.225 \leq \pi \cdot 657.225^3 \cdot [(91.450 - 69.247) + .500 \cdot 91.450]$$

$$= 0.14679 + 06 \leq 0.60582 + 06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-Tube Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(91.450, 0.500 \cdot 91.450 + 91.450 - 146787.0 / (\pi \cdot 657.225^3))$$

$$= \min(91.450, 120.717)$$

$$= 91.450 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [N3-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(2294) + 4 \cdot \text{abs}(0)350.901 \leq \pi \cdot 350.901^3 \cdot [(91.450 - 87.393) + .500 \cdot 91.450]$$

$$= 0.36697 + 05 \leq 0.67574 + 05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Shell Inlet [Pmax]:



$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(91.450, 0.500 \cdot 91.450 + 91.450 - 36696.8 / (\pi \cdot 350.901^3)) \\
 &= \min(91.450, 110.141) \\
 &= 91.450 \text{ kgf/cm}^2
 \end{aligned}$$

Evaluating Flange on Nozzle [N4-Shell Outlet]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(2294) + 4 \cdot \text{abs}(0) 350.901 \leq \pi \cdot 350.901^3 [(91.450 - 87.497) + 0.500 \cdot 91.450] \\
 &= 0.36697 + 05 \leq 0.67433 + 05 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell Outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(91.450, 0.500 \cdot 91.450 + 91.450 - 36696.8 / (\pi \cdot 350.901^3)) \\
 &= \min(91.450, 110.141) \\
 &= 91.450 \text{ kgf/cm}^2
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2020 Metric Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	350	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 6.432

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50 • 1 • 1 • 1 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1984, 1 • 0.98 • 1 • 0.6 • 57.5²)

= 1951.8 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.33 • 0.83 • 1952

= 531.3 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	4.70	1.0000	1.0000	1.0000	57.50	0.8286
20	4.70	1.0000	1.0000	1.0000	57.50	0.8286
30	4.70	1.0000	1.0000	1.0000	57.50	0.8286
40	4.70	1.0000	1.0000	1.0000	57.50	0.8286
50	4.70	1.0000	1.0000	1.0000	57.50	0.8286
60	4.70	1.0000	1.0000	1.0000	57.50	0.8286

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	470	134.16	3285.03	199.115	59.5998
20	30	470	133.2	18648	199.115	338.328
30	40	470	126	4095	199.115	74.295
40	50	470	144	4752	199.115	86.2149
50	60	470	153.12	75028.8	199.115	1361.24
60	70	470	152.64	4506.78	199.115	81.7658

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite 27 Licensee: SL User
FileName : E-30003-AK
Wind Load Calculation:

Step: 15 4:49pm May 21,2026

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HEXAGON

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.094 \cdot 1.4) \cdot 1$
 $= 0.131$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
2.800	0.025
0.000	0.000

Actual Time Period = 0.03 secs
Computed Sa/g = 0.094

Note:

It was indicated to apply the damping factor, this assumes that the Site Spectrum specified is for 5% damping. If your Sa/g values are already per the actual damping value of the vessel then do not apply the damping factor.

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 3763 \cdot 0.13 \cdot 1$
 $= 494.5 \text{ kgf}$

Vertical load per 8.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.13 \cdot 30104 \cdot 1$
 $= 2637.1 \text{ kgf}$

Earthquake Load Calculation:



From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf
10	20	52.5	3763.05	494.459
20	30	52.5	3763.05	494.459
30	40	52.5	3763.05	494.459
40	50	52.5	3763.05	494.459
50	Sad1	52.5	3763.05	494.459
Sad1	60	52.5	3763.05	494.459
50	60	52.5	3763.05	494.459
60	70	52.5	3763.05	494.459

The Earthquake Weight does not include the weight of the saddles.

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**Shop/Field Installation Options :**

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	478.033 cm
Center of Gravity of the Liquid	316.875 cm
Center of Gravity of the Insulation	486.404 cm
Center of Gravity of the Nozzles	169.487 cm
Center of Gravity of the Tubesheet(s)	188.965 cm
Center of Gravity of the Tubes	422.065 cm
Center of Gravity of the Baffles	480.514 cm
Center of Gravity of Bare Shell New and Cold	325.056 cm
Center of Gravity of Bare Shell Corroded	322.308 cm
Vessel CG in the Operating Condition	333.887 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	332.980 cm
Vessel CG in the Test Condition	334.691 cm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	544.00	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	528.08	cm
Distance from Saddle to Vessel tangent	a1 or a	980.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		1014.67	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		24937.90	kgf
Pressure used in Analysis	P	0.000	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-25.99	-2386.14
Long. Stress at Top of Midspan	25.99	1407.13
Long. Stress at Bottom of Midspan	25.99	1407.13
Long. Stress at Top of Saddles	101.86	1407.13
Long. Stress at Bottom of Saddles	-56.46	-2386.14
Long. Stress at Bottom of Saddles	56.46	1407.13
Tangential Shear in Shell	88.81	1125.70
Circ. Stress at Horn of Saddle	95.01	2110.69
Circ. Compressive Stress in Shell	8.27	1407.13

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fw]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 1015 • 1.5 • 1200/2940
 = 621.2 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fw, Fwt, Fsl, Fst)
 = 24317 + max(621, 0, 0, 0)
 = 24937.9 kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	25203.88	kgf
Transverse Shear Load	0.00	kgf
Longitudinal Shear Load	1522.01	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1



Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1050/2 + 38 + 16)\sin(120/2), 2 \cdot 544)$$

$$= 1002.857 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)]$$

$$= -24938 \cdot 98 [1 - (1 - 98/528 + (54.4^2 - 0^2)/(2 \cdot 98 \cdot 528))/(1 + (4 \cdot 0)/(3 \cdot 528))]$$

$$= -3836.6 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [1 + 2(R_m^2 - h_m^2)/(L^2)]/(1 + (4h_m^2)/3L) - 4a/L$$

$$= 24938 \cdot 528/4 [1 + 2(54.4^2 - 0^2)/(528^2)]/(1 + (4 \cdot 0)/(3 \cdot 528)) - 4 \cdot 98/528$$

$$= 9182.6 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 544/(2 \cdot 38) - 9183/(\pi \cdot 544^2 \cdot 38)$$

$$= -25.99 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 544/(2 \cdot 38) + 9183/(\pi \cdot 544^2 \cdot 38)$$

$$= 25.99 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 544/(2 \cdot 38) - 3837/(0.11 \cdot \pi \cdot 544^2 \cdot 38)$$

$$= 101.86 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 544/(2 \cdot 38) + 3837/(0.19 \cdot \pi \cdot 544^2 \cdot 38)$$

$$= -56.46 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 24938(528 - 2 \cdot 98)/(528 + (4 \cdot 0/3))$$

$$= 15682.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 15682 / (544 \cdot 38)$$

$$= 88.81 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(544 \cdot 38)}$$

$$= 112.147 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(544 \cdot 38)}, 2 \cdot 98)$$

$$= 424.29 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$



Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 24938 \cdot 0.1 / (424(38 + 1 \cdot 16))$$

$$= -8.27 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -24938 / (4(38 + 1 \cdot 16)424) - 3 \cdot 0.053 \cdot 24938 / (2(38 + 1 \cdot 16)^2)$$

$$= -95.01 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1044.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2349.3	kgf
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(525 \cdot 38)} + 425)38^3) / 12$$

$$= 295.093 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 295 + 245(175 - 19)^2$$

$$= 60117.500 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	645.3	38.0	19.0	245.2	465936.4	0.601E+05
Wearplate	425.0	16.0	46.0	68.0	312800.0	0.114E+05
Web	10.0	596.0	352.0	59.6	2097919.5	0.363E+05
BasePlate	300.0	25.0	662.5	75.0	4968749.5	0.178E+06
Totals	...	...	...	447.8	7845405.5	0.286E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7845405 / 448$$

$$= 175.187 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 24938$$

$$= 5075.4028 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 25.0514 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1409.5955 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$



$$= 1200 - 525 \cdot \sin(120/2)/1.05$$

$$= 765.828 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 3886.8928 \text{ kgf-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 23.8172 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 1566.2173 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(24938 + 266)300 / (4 \cdot 1044 \cdot 1566)}$$

$$= 18.623 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (525 + 38 + 16)$$

$$= 1002.857 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1003 / (4 - 1)$$

$$= 334.286 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 334 \cdot 300 / 2$$

$$= 501.429 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 25204 / (1044 \cdot 300)$$

$$= 8.047 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 501 \cdot 8.05$$

$$= 4035.104 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19 + 16.7$$

$$= 35.714 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 4035 / 35.7$$

$$= 112.983 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 10$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 35.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$



$$= 200/2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{667/35.7}$$

$$= 43.205 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1200 - \cos(120/2) (525 + 38 + 16) - 25$$

$$= 885.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 1984060 / 2349}$$

$$= 129.113$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 886/43.2$$

$$= 20.495$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 1522 / (2 \cdot 1044) \cdot 334 \cdot 886/2$$

$$= 107.885 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($20.5 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (20.5)^2 / (2 \cdot 129^2)) 2349 / (5/3 + 3 \cdot (20.5) / (8 \cdot 129) - (20.5)^3 / (8 \cdot 129^3))$$

$$= 1344 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 113/1344 + (108 \cdot 100/6666666) / 1566$$

$$= 0.187$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	334.3	10.0	0.0	33.4	0.0	2.79
Totals	...	...	...	52.4	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/52.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1200 - \sqrt{(579 + 16)^2 - (334/2)^2} - 25$$

$$= 620.650 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{669/52.4}$$

$$= 35.731 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



$$= 1 \cdot 621/35.7$$

$$= 17.370$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1522/(2 \cdot 1044)$$

$$= 0.729 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.73 \cdot 334 \cdot 621$$

$$= 151.234 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 151/66937$$

$$= 225.936 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 129^2)) 2349 / (5/3 + 3 \cdot (17.4) / (8 \cdot 129) - (17.4^3) / (8 \cdot 129^3))$$

$$= 1356 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 135/1356 + (151 \cdot 100/669) / 1566$$

$$= 0.244$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1407.13	kgf/cm ²
Area Available in a Single Bolt	BlArea	4.2350	cm ²
Saddle Load Q0 (Weight)	Q0	25203.9	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1522.0	kgf
Maximum Transverse Force	Ft	0.0	kgf
Maximum Longitudinal Force	Fl	1522.0	kgf
Saddle Bolted to Steel Foundation	No		

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1522/(4.23 \cdot 4)$$

$$= 89.8 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0/(4.23 \cdot 4)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 1522/(1100 \cdot 4)$$

$$= 0.3459 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$



$$= 1200 \cdot 0 + 0$$
$$= 0.00 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{\text{mom}} / QO$$
$$= 0/25204$$
$$= 0.00 \text{ mm} < B_{\text{plen}}/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$
$$= \max[0, 0.35, 0]$$
$$= 0.3459 \text{ cm}^2$$

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	545.59	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	528.08	cm
Distance from Saddle to Vessel tangent	a1 or a	980.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		1014.67	kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		24582.47	kgf
Pressure used in Analysis	P	0.000	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-27.81	-2386.14
Long. Stress at Top of Midspan	27.81	1407.13
Long. Stress at Bottom of Midspan	27.81	1407.13
Long. Stress at Top of Saddles	108.81	1407.13
Long. Stress at Bottom of Saddles	-60.31	-2386.14
Long. Stress at Bottom of Saddles	60.31	1407.13
Tangential Shear in Shell	95.25	1125.70
Circ. Stress at Horn of Saddle	104.58	2110.69
Circ. Compressive Stress in Shell	8.86	1407.13

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fw]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 1015 • 1.5 • 1200/2940
 = 621.2 kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fw, Fwt, Fsl, Fst)
 = 23961 + max(621, 0, 0, 0)
 = 24582.5 kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	24848.44	kgf
Transverse Shear Load	0.00	kgf
Longitudinal Shear Load	1522.01	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1



Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1050/2 + 38 + 16)\sin(120/2), 2 \cdot 546)$$

$$= 1002.857 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)]$$

$$= -24582 \cdot 98 [1 - (1 - 98/528 + (54.6^2 - 0^2)/(2 \cdot 98 \cdot 528))/(1 + (4 \cdot 0)/(3 \cdot 528))]$$

$$= -3777.9 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [1 + 2(R_m^2 - h_m^2)/(L^2)]/(1 + (4h_m^2)/3L) - 4a/L$$

$$= 24582 \cdot 528/4 [1 + 2(54.6^2 - 0^2)/(528^2)]/(1 + (4 \cdot 0)/(3 \cdot 528)) - 4 \cdot 98/528$$

$$= 9055.8 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 546/(2 \cdot 34.8) - 9056/(\pi \cdot 546^2 \cdot 34.8)$$

$$= -27.81 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 546/(2 \cdot 34.8) + 9056/(\pi \cdot 546^2 \cdot 34.8)$$

$$= 27.81 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 546/(2 \cdot 34.8) - 3778/(0.11 \cdot \pi \cdot 546^2 \cdot 34.8)$$

$$= 108.81 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 546/(2 \cdot 34.8) + 3778/(0.19 \cdot \pi \cdot 546^2 \cdot 34.8)$$

$$= -60.31 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 24582(528 - 2 \cdot 98)/(528 + (4 \cdot 0/3))$$

$$= 15458.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 15459 / (546 \cdot 34.8)$$

$$= 95.25 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(546 \cdot 34.8)}$$

$$= 107.516 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(546 \cdot 34.8)}, 2 \cdot 98)$$

$$= 415.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$



Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 24582 \cdot 0.1 / (415(34.8 + 1 \cdot 16))$$

$$= -8.86 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -24582 / (4(34.8 + 1 \cdot 16)415) -$$

$$3 \cdot 0.053 \cdot 24582 / (2(34.8 + 1 \cdot 16)^2)$$

$$= -104.58 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1044.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2349.3	kgf
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(528 \cdot 34.8)} + 425)34.8^3) / 12$$

$$= 224.047 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 224 + 222(182 - 17.4)^2$$

$$= 59918.535 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	636.6	34.8	17.4	221.7	386011.3	0.599E+05
Wearplate	425.0	16.0	42.8	68.0	291210.0	0.131E+05
Web	10.0	596.0	348.8	59.6	2078996.5	0.343E+05
BasePlate	300.0	25.0	659.3	75.0	4944937.0	0.171E+06
Totals	...	...	...	424.3	7701155.0	0.279E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7701155 / 424$$

$$= 181.508 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 24582$$

$$= 5003.0649 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 24.6943 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1409.5955 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$



$$= 1200 - 528 \cdot \sin(120/2)/1.05$$

$$= 763.203 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 3818.3577 \text{ kgf-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 24.8758 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 1566.2173 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(24582 + 266)300 / (4 \cdot 1044 \cdot 1566)}$$

$$= 18.491 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2)(\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2)(525 + 38 + 16)$$

$$= 1002.857 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1003 / (4 - 1)$$

$$= 334.286 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 334 \cdot 300 / 2$$

$$= 501.429 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 24848 / (1044 \cdot 300)$$

$$= 7.934 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 501 \cdot 7.93$$

$$= 3978.200 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19 + 16.7$$

$$= 35.714 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 3978 / 35.7$$

$$= 111.390 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 10$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 35.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$



$$= 200/2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{667/35.7}$$

$$= 43.205 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1200 - \cos(120/2) (525 + 38 + 16) - 25$$

$$= 885.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 1984060 / 2349}$$

$$= 129.113$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 886 / 43.2$$

$$= 20.495$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 1522 / (2 \cdot 1044) \cdot 334 \cdot 886 / 2$$

$$= 107.885 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($20.5 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (20.5)^2 / (2 \cdot 129^2)) 2349 / (5/3 + 3 \cdot (20.5) / (8 \cdot 129) - (20.5)^3 / (8 \cdot 129^3))$$

$$= 1344 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 111/1344 + (108 \cdot 100/6666666) / 1566$$

$$= 0.186$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	334.3	10.0	0.0	33.4	0.0	2.79
Totals	...	...	...	52.4	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/52.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1200 - \sqrt{(579 + 16)^2 - (334/2)^2} - 25$$

$$= 620.650 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{669/52.4}$$

$$= 35.731 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



$$= 1 \cdot 621/35.7$$

$$= 17.370$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1522/(2 \cdot 1044)$$

$$= 0.729 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.73 \cdot 334 \cdot 621$$

$$= 151.234 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 151/66937$$

$$= 225.936 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 129^2)) 2349 / (5/3 + 3 \cdot (17.4) / (8 \cdot 129) - (17.4^3) / (8 \cdot 129^3))$$

$$= 1356 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 133/1356 + (151 \cdot 100/669) / 1566$$

$$= 0.242$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1407.13	kgf/cm ²
Area Available in a Single Bolt	BlArea	4.2350	cm ²
Saddle Load Q0 (Weight)	Q0	24848.4	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1522.0	kgf
Maximum Transverse Force	Ft	0.0	kgf
Maximum Longitudinal Force	Fl	1522.0	kgf
Saddle Bolted to Steel Foundation	No		

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1522/(4.23 \cdot 4)$$

$$= 89.8 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0/(4.23 \cdot 4)$$

$$= 0.0 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 1522/(1100 \cdot 4)$$

$$= 0.3459 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$



$$= 1200 \cdot 0 + 0$$
$$= 0.00 \text{ kgf-m.}$$

Eccentricity (e):
= Rmom / QO
= 0/24848
= 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.35, 0]
= 0.3459 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	545.59	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	528.08	cm
Distance from Saddle to Vessel tangent	a1 or a	980.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		38491.39	kgf
Pressure used in Analysis	P	85.547	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	626.57	1407.13
Long. Stress at Bottom of Midspan	713.65	1407.13
Long. Stress at Top of Saddles	840.49	1407.13
Long. Stress at Bottom of Saddles	575.68	1407.13
Tangential Shear in Shell	149.14	1125.70
Circ. Stress at Horn of Saddle	163.75	1758.91
Circ. Compressive Stress in Shell	13.87	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2001/2 + 0) \cdot 1200/1003$
 $= 3592.3 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(314, 0) \cdot 1200/2940$
 $= 128.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(3956, 0) \cdot 1200/2940$
 $= 1614.6 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (3956/2 + 0) \cdot 1200/1003$
 $= 7099.9 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$



$$= 31391 + \max(128, 3592, 1615, 7100)$$

$$= 38491.4 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1953(0 + \pi/4(1.13 \cdot 1.2)^2)$$

$$= 3079.973 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	38757.37	kgf
Transverse Shear Load	1977.84	kgf
Longitudinal Shear Load	3955.67	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1050/2 + 38 + 16)\sin(120/2), 2 \cdot 546)$$

$$= 1002.857 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)]$$

$$= -38491 \cdot 98 [1 - (1 - 98/528 + (54.6^2 - 0^2)/(2 \cdot 98 \cdot 528))/(1 + (4 \cdot 0)/(3 \cdot 528))]$$

$$= -5915.5 \text{ kgf-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L$$

$$= 38491 \cdot 528/4(1 + 2(54.6^2 - 0^2)/(528^2))/(1 + (4 \cdot 0)/(3 \cdot 528)) - 4 \cdot 98/528$$

$$= 14179.6 \text{ kgf-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 85.5 \cdot 546/(2 \cdot 34.8) - 14180/(\pi \cdot 546^2 \cdot 34.8)$$

$$= 626.57 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 85.5 \cdot 546/(2 \cdot 34.8) + 14180/(\pi \cdot 546^2 \cdot 34.8)$$

$$= 713.65 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 85.5 \cdot 546/(2 \cdot 34.8) - 5915/(0.11 \cdot \pi \cdot 546^2 \cdot 34.8)$$

$$= 840.49 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 85.5 \cdot 546/(2 \cdot 34.8) + 5915/(0.19 \cdot \pi \cdot 546^2 \cdot 34.8)$$

$$= 575.68 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 38491(528 - 2 \cdot 98)/(528 + (4 \cdot 0/3))$$

$$= 24205.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 24205 / (546 \cdot 34.8)$$



= 149.14 kgf/cm²

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(546 \cdot 34.8)}$$

$$= 107.516 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(546 \cdot 34.8)}, 2 \cdot 98)$$

$$= 415.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(1407/1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_{6,r}]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 38491 \cdot 0.1 / (415(34.8 + 1 \cdot 16))$$

$$= -13.87 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.28) [σ_{7,r}]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -38491 / (4(34.8 + 1 \cdot 16)415) -$$

$$3 \cdot 0.053 \cdot 38491 / (2(34.8 + 1 \cdot 16)^2)$$

$$= -163.75 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 2940.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12491\text{E-}04 \cdot 2940(160 - 21.1)$$

$$= 5.100 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1044.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2349.3	kgf
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(528 \cdot 34.8)} + 425)34.8^3)/12$$

$$= 224.047 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 224 + 222(182 - 17.4)^2$$

$$= 59918.535 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	636.6	34.8	17.4	221.7	386011.3	0.599E+05
Wearplate	425.0	16.0	42.8	68.0	291210.0	0.131E+05
Web	10.0	596.0	348.8	59.6	2078996.5	0.343E+05
BasePlate	300.0	25.0	659.3	75.0	4944937.0	0.171E+06
Totals	...	...	...	424.3	7701155.0	0.279E+06

Distance to Centroid [C1]:



$$\begin{aligned}
 &= AY / A \\
 &= 7701155/424 \\
 &= 181.508 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.2 \cdot 38491 \\
 &= 7833.8335 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 38.6666 \text{ kgf/cm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1409.5955 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1200 - 528 \cdot \sin(120/2) / 1.05 \\
 &= 763.203 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5978.8101 \text{ kgf-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 38.9508 \text{ kgf/cm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1566.2173 \text{ kgf/cm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(38491 + 266)300 / (4 \cdot 1044 \cdot 1566)} \\
 &= 23.094 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (525 + 38 + 16) \\
 &= 1002.857 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 1003 / (4 - 1) \\
 &= 334.286 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot \text{Bpwid} / 2 \\
 &= 334 \cdot 300/2 \\
 &= 501.429 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 38757 / (1044 \cdot 300) \\
 &= 12.375 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 501 \cdot 12.4 \\
 &= 6205.000 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of the Rib and Web [Ar]:} \\
 &= \text{Rib Area} + \text{Web Area} \\
 &= 19 + 16.7 \\
 &= 35.714 \text{ cm}^2
 \end{aligned}$$

$$\text{Compressive Stress [Sc]:}$$



$$\begin{aligned}
 &= P/Ar \\
 &= 6205/35.7 \\
 &= 173.740 \text{ kgf/cm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 200 - 10 \\
 &= 190.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0/35.7 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 200/2 \\
 &= 100.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(667/35.7)} \\
 &= 43.205 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1200 - \cos(120/2) (525 + 38 + 16) - 25 \\
 &= 885.500 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 1984060/2349)} \\
 &= 129.113
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 886/43.2 \\
 &= 20.495
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2 \\
 &= 3956 / (2 \cdot 1044) \cdot 334 \cdot 886/2 \\
 &= 280.393 \text{ kgf-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($20.5 < 129$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (20.5)^2 / (2 \cdot 129^2)) 2349 / \\
 &\quad (5/3 + 3 \cdot (20.5) / (8 \cdot 129) - (20.5)^3 / (8 \cdot 129^3)) \\
 &= 1344 \text{ kgf/cm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\
 &= 174/1344 + (280 \cdot 100/6666666) / 1566 \\
 &= 0.398
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	334.3	10.0	0.0	33.4	0.0	2.79



Totals	...	...	...	52.4	...	669.
--------	-----	-----	-----	------	-----	------

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/52.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1200 - \sqrt{(579 + 16)^2 - (334/2)^2} - 25$$

$$= 620.650 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(669/52.4)}$$

$$= 35.731 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 621/35.7$$

$$= 17.370$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3956 / (2 \cdot 1044)$$

$$= 1.894 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.89 \cdot 334 \cdot 621$$

$$= 393.056 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 393/66937$$

$$= 587.204 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 129^2)) 2349 /$$

$$(5/3 + 3 \cdot (17.4) / (8 \cdot 129) - (17.4)^3 / (8 \cdot 129^3))$$

$$= 1356 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 207/1356 + (393 \cdot 100/669) / 1566$$

$$= 0.528$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1407.13	kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350	cm ²
Saddle Load Q0 (Weight)	Q0	31657.5	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1614.6	kgf
Maximum Transverse Force	Ft	1977.8	kgf
Maximum Longitudinal Force	F1	3955.7	kgf
Saddle Bolted to Steel Foundation	No		



Friction force due to thermal growth, [Ff]:
 = 484.8 kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $\max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $3956 / (4.23 \cdot 4)$
 = 233.5 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $1978 / (4.23 \cdot 4)$
 = 116.8 kgf/cm². Must be less than 1100.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $3956 / (1100 \cdot 4)$
 = 0.8990 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1200 \cdot 1978 + 0$
 = 2373.41 kgf-m.

Eccentricity (e):
 = R_{mom} / QO
 = $2373 / 31657$
 = 74.97 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 0.9, 0]$
 = 0.8990 cm²

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	545.59	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	528.08	cm
Distance from Saddle to Vessel tangent	a1 or a	980.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.21	cm
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	



Distance from Saddle Base to Centerline B 1200.00 mm
 Coefficient of Friction mu 0.30

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 8449.94 kgf
 Pressure used in Analysis P 85.547 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	663.06	1407.13
Long. Stress at Bottom of Midspan	677.16	1407.13
Long. Stress at Top of Saddles	722.60	1407.13
Long. Stress at Bottom of Saddles	641.02	1407.13
Tangential Shear in Shell	30.56	1125.70
Circ. Stress at Horn of Saddle	35.95	1758.91
Circ. Compressive Stress in Shell	3.05	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (2001/2 + 0) \cdot 1200/1003$
 $= 3592.3\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(314, 0) \cdot 1200/2940$
 $= 128.2\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(3956, 0) \cdot 1200/2940$
 $= 1614.6\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (3956/2 + 0) \cdot 1200/1003$
 $= 7099.9\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 1350 + \max(128, 3592, 1615, 7100)$
 $= 8449.9\ kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1953 (0 + \pi/4(1.13 \cdot 1.2)^2)$
 $= 3079.973\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 8715.91 kgf
 Transverse Shear Load 1977.84 kgf
 Longitudinal Shear Load 3955.67 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1050/2 + 38 + 16)\sin(120/2), 2 \cdot 546)$
 $= 1002.857\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434



The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4 h m^2)}{3L} \right) \right]$$

$$= -8450 \cdot 98 \left[1 - \left(1 - \frac{98}{528} + \frac{(54.6^2 - 28.2^2)}{(2 \cdot 98 \cdot 528)} \right) / \left(1 + \frac{(4 \cdot 28.2)}{(3 \cdot 528)} \right) \right]$$

$$= -1822.3 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(1 + 2 \frac{(R_m^2 - h m^2)}{(L^2)} \right) / \left(1 + \frac{(4 h m^2)}{3L} \right) - 4a/L$$

$$= 8450 \cdot 528 / 4 \left(1 + 2 \frac{(54.6^2 - 28.2^2)}{(528^2)} \right) / \left(1 + \frac{(4 \cdot 28.2)}{(3 \cdot 528)} \right) - 4 \cdot 98 / 528$$

$$= 2295.8 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 85.5 \cdot 546 / (2 \cdot 34.8) - 2296 / (\pi \cdot 546^2 \cdot 34.8)$$

$$= 663.06 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 85.5 \cdot 546 / (2 \cdot 34.8) + 2296 / (\pi \cdot 546^2 \cdot 34.8)$$

$$= 677.16 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 85.5 \cdot 546 / (2 \cdot 34.8) - 1822 / (0.11 \cdot \pi \cdot 546^2 \cdot 34.8)$$

$$= 722.60 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 85.5 \cdot 546 / (2 \cdot 34.8) + 1822 / (0.19 \cdot \pi \cdot 546^2 \cdot 34.8)$$

$$= 641.02 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h m / 3))$$

$$= 8450 (528 - 2 \cdot 98) / (528 + (4 \cdot 28.2 / 3))$$

$$= 4960.4 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 4960 / (546 \cdot 34.8)$$

$$= 30.56 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(546 \cdot 34.8)}$$

$$= 107.516 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(546 \cdot 34.8)}, 2 \cdot 98)$$

$$= 415.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(1407/1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 8450 \cdot 0.1 / (415(34.8 + 1 \cdot 16))$$

$$= -3.05 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -8450 / (4(34.8 + 1 \cdot 16)415) -$$



$$3 \cdot 0.053 \cdot 8450 / (2(34.8 + 1 \cdot 16)^2) \\ = -35.95 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1044.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2349.3	kgf
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12 \\ = ((1.56 \cdot \sqrt{(528 \cdot 34.8)} + 425) 34.8^3) / 12 \\ = 224.047 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2 \\ = 224 + 222 (182 - 17.4)^2 \\ = 59918.535 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	636.6	34.8	17.4	221.7	386011.3	0.599E+05
Wearplate	425.0	16.0	42.8	68.0	291210.0	0.131E+05
Web	10.0	596.0	348.8	59.6	2078996.5	0.343E+05
BasePlate	300.0	25.0	659.3	75.0	4944937.0	0.171E+06
Totals	...	...	...	424.3	7701155.0	0.279E+06

Distance to Centroid [C1]:

$$= AY / A \\ = 7701155 / 424 \\ = 181.508 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2 \\ = 180 - 120 / 2 \\ = 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ = (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ = 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q \\ = 0.2 \cdot 8450 \\ = 1719.7455 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 8.4884 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1409.5955 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ = 1200 - 528 \cdot \sin(120/2) / 1.05 \\ = 763.203 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1312.5160 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 8.5508 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1566.2173 \text{ kgf/cm}^2$$



Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{3(8450 + 266)300 / (4 \cdot 1044 \cdot 1566)} \\ &= 10.951 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (525 + 38 + 16) \\ &= 1002.857 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1003 / (4 - 1) \\ &= 334.286 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 334 \cdot 300 / 2 \\ &= 501.429 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 8716 / (1044 \cdot 300) \\ &= 2.783 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= \text{Ap} \cdot \text{Bp} \\ &= 501 \cdot 2.78 \\ &= 1395.405 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 19 + 16.7 \\ &= 35.714 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / \text{Ar} \\ &= 1395 / 35.7 \\ &= 39.071 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$\begin{aligned} &= \text{Saddle Width} - \text{Web Thickness} \\ &= 200 - 10 \\ &= 190.000 \text{ mm} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= \text{AY} / \text{A} \\ &= 0 / 35.7 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 200 / 2 \\ &= 100.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{667 / 35.7} \\ &= 43.205 \text{ mm} \end{aligned}$$



Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1200 - \cos(120/2) (525 + 38 + 16) - 25$
 $= 885.500 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 1984060 / 2349}$
 $= 129.113$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 886 / 43.2$
 $= 20.495$

Bending Moment [Rm]:
 $= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 $= 3956 / (2 \cdot 1044) \cdot 334 \cdot 886 / 2$
 $= 280.393 \text{ kgf-m.}$

Compressive Allowable, $KL/r < Cc$ ($20.5 < 129$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (20.5)^2 / (2 \cdot 129^2)) 2349 /$
 $(5/3 + 3 \cdot (20.5) / (8 \cdot 129) - (20.5^3) / (8 \cdot 129^3))$
 $= 1344 \text{ kgf/cm}^2$

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 39.1 / 1344 + (280 \cdot 100 / 6666666) / 1566$
 $= 0.298$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	334.3	10.0	0.0	33.4	0.0	2.79
Totals	...	...	...	52.4	...	669.

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 52.4$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 200 / 2$
 $= 100.000 \text{ mm}$

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$
 $= 1200 - \sqrt{(579 + 16)^2 - (334/2)^2} - 25$
 $= 620.650 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{669 / 52.4}$
 $= 35.731 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 621 / 35.7$
 $= 17.370$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 3956 / (2 \cdot 1044)$
 $= 1.894 \text{ kgf/mm}$



Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.89 \cdot 334 \cdot 621$$

$$= 393.056 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 393/66937$$

$$= 587.204 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 129^2)) 2349 / (5/3 + 3 \cdot (17.4) / (8 \cdot 129) - (17.4^3) / (8 \cdot 129^3))$$

$$= 1356 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 46.6/1356 + (393 \cdot 100/669) / 1566$$

$$= 0.409$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1407.13	kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350	cm ²
Saddle Load QO (Weight)	QO	1616.0	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1614.6	kgf
Maximum Transverse Force	Ft	1977.8	kgf
Maximum Longitudinal Force	Fl	3955.7	kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3956 / (4.23 \cdot 4)$$

$$= 233.5 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1978 / (4.23 \cdot 4)$$

$$= 116.8 \text{ kgf/cm}^2. \text{ Must be less than } 1100.0 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 3956 / (1100 \cdot 4)$$

$$= 0.8990 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1200 \cdot 1978 + 0$$

$$= 2373.41 \text{ kgf-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 2373/1616$$

$$= 1468.69 \text{ mm} > B_{plen}/6 \text{ --> Uplift in Transverse direction}$$



$$\begin{aligned}
 f &= B_{plen} / 2 - \text{Edgedis} \\
 &= 1044/2 - 50.8 \\
 &= 471.20 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Modular Ratio Of Steel/Concrete (n1):} \\
 &= E_S / E_C \\
 &= 2074056/219500 \\
 &= 9.45
 \end{aligned}$$

$$\begin{aligned}
 K1 &= 3 (e - 0.5 \cdot B_{plen}) \\
 &= 3 (1469 - 0.5 \cdot 1044) \\
 &= 2840.08 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 K2 &= 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e) \\
 &= 6 \cdot 9.45 \cdot 8.47/300 \cdot (471 + 1469) \\
 &= 310508.08 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 K3 &= -K2 \cdot (0.5 \cdot B_{plen} + f) \\
 &= -310508 \cdot (0.5 \cdot 1044 + 471) \\
 &= -308396624.77 \text{ mm}^3
 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned}
 Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\
 Y^3 + 2840 \cdot Y^2 + 310508 \cdot Y + -308396608 &= 0 \\
 Y &= 268.95 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Num} &= (B_{plen} / 2 - Y / 3 - e) \\
 &= (1044/2 - 269/3 - 1469) \\
 &= -1036.34
 \end{aligned}$$

$$\begin{aligned}
 \text{Denom} &= (B_{plen} / 2 - Y / 3 + f) \\
 &= (1044/2 - 269/3 + 471) \\
 &= 903.55
 \end{aligned}$$

$$\begin{aligned}
 \text{Total Bolt Tension Force [Tforce]:} \\
 &= -QO \cdot \text{Num} / \text{Denom} \\
 &= -1616 \cdot -1036/904 \\
 &= 1853.49 \text{ kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Bolt Area Required due to Transverse Load [Bltareart]:} \\
 &= T_{force} / (S_{tba} \cdot N_{bt}) \\
 &= 1853/(1400 \cdot 2) \\
 &= 0.6620 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Required Area of a Single Bolt [Bltarear]:} \\
 &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\
 &= \max[0, 0.9, 0.66] \\
 &= 0.8990 \text{ cm}^2
 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned}
 \text{Bearing Pressure (fc)} \\
 &= 2(QO + T_{force}) / (Y \cdot B_{pwid}) \\
 &= 2(1616 + 1853)/(269 \cdot 300) \\
 &= 8.60 \text{ kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance from Baseplate Edge to the Web [ADIST]:} \\
 &= (B_{plen} - \text{WebLength}) / 2 \\
 &= (1044 - 993)/2 \\
 &= 25.4000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Overturning Moment due To Bolt Tension [Mt]:} \\
 &= T_{force} \cdot \text{Adist} \\
 &= 1853 \cdot 2.54 \\
 &= 47.08 \text{ kgf-m.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Equivalent Bearing Pressure (f1):} \\
 &= f_c \cdot (Y - \text{Adist}) / Y \\
 &= 8.6 \cdot (269 - 25.4)/269
 \end{aligned}$$



$$= 7.79 \text{ kgf/cm}^2$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$$

$$= (25.4^2 \cdot 300 / 6) \cdot (7.79 + 2 \cdot 8.6)$$

$$= 8.06 \text{ kgf-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$$

$$= \sqrt{(6 \cdot \max(47.1, 8.06) / (300 \cdot 2111))}$$

$$= 6.6791 \text{ mm}$$

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	544.00	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	528.08	cm
Distance from Saddle to Vessel tangent	a1 or a	980.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		2386.14	kgf/cm ²
Head Allowable Stress used in Calculation		2386.14	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	P	31913.72	kgf
Pressure used in Analysis	P	124.531	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	858.11	2386.14
Long. Stress at Bottom of Midspan	924.64	2386.14
Long. Stress at Top of Saddles	1021.73	2386.14
Long. Stress at Bottom of Saddles	819.13	2386.14
Tangential Shear in Shell	113.65	1413.13
Circ. Stress at Horn of Saddle	121.59	3579.21
Circ. Compressive Stress in Shell	10.59	2386.14

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (660/2 + 0) \cdot 1200/1003$
 $= 1185.5 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(104, 0) \cdot 1200/2940$
 $= 42.3 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 30728 + \max(42.3, 1185, 0, 0)$
 $= 31913.7 \text{ kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1953 (0 + \pi/4 (1.13 \cdot 1.2)^2)$
 $= 3079.973 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 32179.70 kgf



Transverse Shear Load 330.24 kgf
 Longitudinal Shear Load 103.64 kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1050/2 + 38 + 16)\sin(120/2), 2 \cdot 544)$$

$$= 1002.857 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)]$$

$$= -31914 \cdot 98 [1 - (1 - 98/528 + (54.4^2 - 0^2)/(2 \cdot 98 \cdot 528))/(1 + (4 \cdot 0)/(3 \cdot 528))]$$

$$= -4909.8 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/3L) - 4a/L$$

$$= 31914 \cdot 528/4(1 + 2(54.4^2 - 0^2)/(528^2))/(1 + (4 \cdot 0)/(3 \cdot 528)) - 4 \cdot 98/528$$

$$= 11751.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 125 \cdot 544/(2 \cdot 38) - 11751/(\pi \cdot 544^2 \cdot 38)$$

$$= 858.11 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 125 \cdot 544/(2 \cdot 38) + 11751/(\pi \cdot 544^2 \cdot 38)$$

$$= 924.64 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 125 \cdot 544/(2 \cdot 38) - 4910/(0.11 \cdot \pi \cdot 544^2 \cdot 38)$$

$$= 1021.73 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 125 \cdot 544/(2 \cdot 38) + 4910/(0.19 \cdot \pi \cdot 544^2 \cdot 38)$$

$$= 819.13 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 31914(528 - 2 \cdot 98)/(528 + (4 \cdot 0/3))$$

$$= 20068.8 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 20069 / (544 \cdot 38)$$

$$= 113.65 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{544 \cdot 38}$$

$$= 112.147 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{544 \cdot 38}, 2 \cdot 98)$$



= 424.29 mm

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 = 1.0000 Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 = $-K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 = $-0.76 \cdot 31914 \cdot 0.1 / (424(38 + 1 \cdot 16))$
 = -10.59 kgf/cm^2

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:
 = $-Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$
 = $-31914 / (4(38 + 1 \cdot 16)424) -$
 $3 \cdot 0.053 \cdot 31914 / (2(38 + 1 \cdot 16)^2)$
 = -121.59 kgf/cm^2

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1044.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2349.3	kgf
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 = $((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$
 = $((1.56 \cdot \sqrt{(525 \cdot 38)} + 425)38^3) / 12$
 = 295.093 cm^4

Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = $295 + 245(175 - 19)^2$
 = 60117.500 cm^4

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	645.3	38.0	19.0	245.2	465936.4	0.601E+05
Wearplate	425.0	16.0	46.0	68.0	312800.0	0.114E+05
Web	10.0	596.0	352.0	59.6	2097919.5	0.363E+05
BasePlate	300.0	25.0	662.5	75.0	4968749.5	0.178E+06
Totals	...	...	...	447.8	7845405.5	0.286E+06

Distance to Centroid [C1]:
 = AY / A
 = $7845405 / 448$
 = 175.187 mm

Angle [β]:
 = $180 - \text{Saddle Angle} / 2$
 = $180 - 120 / 2$
 = 120.0

Saddle Splitting Coefficient [K1]:
 = $(1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 = $(1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$
 = 0.2035

Saddle Splitting Force [Fh]:
 = $K1 \cdot Q$
 = $0.2 \cdot 31914$
 = 6495.1343 kgf

Tension Stress, $St = (Fh / As) = 32.0589 \text{ kgf/cm}^2$



Allowed Stress, $S_a = 0.6 \cdot \text{Yield Str} = 1409.5955 \text{ kgf/cm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1200 - 525 \cdot \sin(120/2) / 1.05$
 $= 765.828 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 4974.1650 \text{ kgf-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 30.4795 \text{ kgf/cm}^2$
Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 1566.2173 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$
 $= \sqrt{\frac{3(31914 + 266) \cdot 300}{4 \cdot 1044 \cdot 1566}}$
 $= 21.043 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (525 + 38 + 16)$
 $= 1002.857 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 1003 / (4 - 1)$
 $= 334.286 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 334 \cdot 300 / 2$
 $= 501.429 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 32180 / (1044 \cdot 300)$
 $= 10.274 \text{ kgf/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 501 \cdot 10.3$
 $= 5151.924 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 19 + 16.7$
 $= 35.714 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 5152 / 35.7$
 $= 144.254 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 200 - 10$
 $= 190.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 35.7$



= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 200/2
 = 100.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{667/35.7}$
 = 43.205 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1200 - cos(120/2)(525 + 38 + 16) - 25
 = 885.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 1984060/2349)}$
 = 129.113

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 886/43.2
 = 20.495

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 104 / (2 • 1044) • 334 • 886/2
 = 7.347 kgf-m.

Compressive Allowable, KL/r < Cc (20.5 < 129) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (20.5)^2 / (2 \cdot 129^2)) 2349 / (5/3 + 3 \cdot (20.5) / (8 \cdot 129) - (20.5)^3 / (8 \cdot 129^3))$
 = 1344 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 144/1344 + (7.35 • 100/6666666) / 1566
 = 0.114

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	334.3	10.0	0.0	33.4	0.0	2.79
Totals	...	...	...	52.4	...	669.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/52.4
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 200/2
 = 100.000 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk
 = 1200 - $\sqrt{(579 + 16)^2 - (334/2)^2}$ - 25
 = 620.650 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{669/52.4}$



= 35.731 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 621/35.7

= 17.370

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 104/(2 • 1044)

= 0.050 kgf/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.05 • 334 • 621

= 10.298 kgf-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 10.3/66937

= 15.385 kgf/cm²

Compressive Allowable, KL/r < Cc (17.4 < 129) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³)/(8•Cc³)

= (1-(17.4)²/(2 • 129²))2349/

(5/3+3•(17.4)/(8• 129)-(17.4³)/(8•129³)

= 1356 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 172/1356 + (10.3 • 100/669)/1566

= 0.137

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1407.13	kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350	cm ²
Saddle Load Q0 (Weight)	Q0	30994.2	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	42.3	kgf
Maximum Transverse Force	Ft	330.2	kgf
Maximum Longitudinal Force	Fl	103.6	kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= FI / (Bolt Area • Number of Bolts)

= 104/(4.23 • 4)

= 6.1 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)

= 330/(4.23 • 4)

= 19.5 kgf/cm². Must be less than 1100.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= FI / (BoltShearAllowable • Nbolts)

= 104/(1100 • 4)

= 0.0236 cm²



Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1200 \cdot 330 + 0$$

$$= 396.29 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 396/30994$$

$$= 12.79 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.024, 0]$$

$$= 0.0236 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	544.00	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	528.08	cm
Distance from Saddle to Vessel tangent	a1 or a	980.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.05	cm
Shell Allowable Stress used in Calculation		2386.14	kgf/cm ²
Head Allowable Stress used in Calculation		2386.14	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		1310.08	kgf
Pressure used in Analysis	P	124.531	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		890.37	2386.14
Long. Stress at Bottom of Midspan		892.39	2386.14
Long. Stress at Top of Saddles		898.87	2386.14
Long. Stress at Bottom of Saddles		887.22	2386.14
Tangential Shear in Shell		4.36	1413.13
Circ. Stress at Horn of Saddle		4.99	3579.21
Circ. Compressive Stress in Shell		0.43	2386.14

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z \text{ Force Load}) \cdot B / E$$



$$= 3 (660/2 + 0) \cdot 1200/1003$$

$$= 1185.5 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(104, 0) \cdot 1200/2940$$

$$= 42.3 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 125 + \max(42.3, 1185, 0, 0)$$

$$= 1310.1 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1953 (0 + \pi/4 (1.13 \cdot 1.2)^2)$$

$$= 3079.973 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1576.06	kgf
Transverse Shear Load	330.24	kgf
Longitudinal Shear Load	103.64	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1050/2 + 38 + 16) \sin(120/2), 2 \cdot 544)$$

$$= 1002.857 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m^2)/3L)]$$

$$= -1310 \cdot 98 [1 - (1 - 98/528 + (54.4^2 - 28.1^2) / (2 \cdot 98 \cdot 528)) / (1 + (4 \cdot 28.1) / (3 \cdot 528))]$$

$$= -282.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 (1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L$$

$$= 1310 \cdot 528 / 4 (1 + 2(54.4^2 - 28.1^2)/(528^2)) / (1 + (4 \cdot 28.1) / (3 \cdot 528)) - 4 \cdot 98/528$$

$$= 356.5 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 125 \cdot 544 / (2 \cdot 38) - 356 / (\pi \cdot 544^2 \cdot 38)$$

$$= 890.37 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 125 \cdot 544 / (2 \cdot 38) + 356 / (\pi \cdot 544^2 \cdot 38)$$

$$= 892.39 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 125 \cdot 544 / (2 \cdot 38) - 282 / (0.11 \cdot \pi \cdot 544^2 \cdot 38)$$

$$= 898.87 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 125 \cdot 544 / (2 \cdot 38) + 282 / (0.19 \cdot \pi \cdot 544^2 \cdot 38)$$



$$= 887.22 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 1310 (528 - 2 \cdot 98)/(528 + (4 \cdot 28.1/3))$$

$$= 769.4 \text{ kgf}$$

Shear Stress in the shell no ribs, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 769 / (544 \cdot 38)$$

$$= 4.36 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(544 \cdot 38)}$$

$$= 112.147 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(544 \cdot 38)}, 2 \cdot 98)$$

$$= 424.29 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 1310 \cdot 0.1 / (424(38 + 1 \cdot 16))$$

$$= -0.43 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q/(4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -1310 / (4(38 + 1 \cdot 16)424) -$$

$$3 \cdot 0.053 \cdot 1310 / (2(38 + 1 \cdot 16)^2)$$

$$= -4.99 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1044.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2349.3	kgf
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(525 \cdot 38)} + 425)38^3)/12$$

$$= 295.093 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 295 + 245(175 - 19)^2$$

$$= 60117.500 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	645.3	38.0	19.0	245.2	465936.4	0.601E+05
Wearplate	425.0	16.0	46.0	68.0	312800.0	0.114E+05
Web	10.0	596.0	352.0	59.6	2097919.5	0.363E+05
BasePlate	300.0	25.0	662.5	75.0	4968749.5	0.178E+06
Totals	...	...	...	447.8	7845405.5	0.286E+06



Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 7845405/448 \\ &= 175.187 \text{ mm} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.2 \cdot 1310 \\ &= 266.6303 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh / As) = 1.3160 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1409.5955 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ &= 1200 - 525 \cdot \sin(120/2) / 1.05 \\ &= 765.828 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 204.1933 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.2512 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1566.2173 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})) } \\ &= \sqrt{ (3(1310 + 266)300 / (4 \cdot 1044 \cdot 1566)) } \\ &= 4.657 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120/2) (525 + 38 + 16) \\ &= 1002.857 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1003 / (4 - 1) \\ &= 334.286 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 334 \cdot 300 / 2 \\ &= 501.429 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 1576 / (1044 \cdot 300) \\ &= 0.503 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 501 \cdot 0.5 \\ &= 252.325 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 19 + 16.7 \\ &= 35.714 \text{ cm}^2 \end{aligned}$$



Compressive Stress [Sc]:

$$= P/A_r$$

$$= 252/35.7$$

$$= 7.065 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 10$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/35.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(667/35.7)}$$

$$= 43.205 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1200 - \cos(120/2) (525 + 38 + 16) - 25$$

$$= 885.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 1984060/2349)}$$

$$= 129.113$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 886/43.2$$

$$= 20.495$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 104 / (2 \cdot 1044) \cdot 334 \cdot 886/2$$

$$= 7.347 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($20.5 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (20.5)^2 / (2 \cdot 129^2)) 2349 /$$

$$(5/3 + 3 \cdot (20.5) / (8 \cdot 129) - (20.5)^3 / (8 \cdot 129^3))$$

$$= 1344 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 7.07/1344 + (7.35 \cdot 100/6666666) / 1566$$

$$= 0.012$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.



Web	334.3	10.0	0.0	33.4	0.0	2.79
Totals	...	...	...	52.4	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/52.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1200 - \sqrt{(579 + 16)^2 - (334/2)^2} - 25$$

$$= 620.650 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(669/52.4)}$$

$$= 35.731 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 621/35.7$$

$$= 17.370$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 104/(2 \cdot 1044)$$

$$= 0.050 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.05 \cdot 334 \cdot 621$$

$$= 10.298 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 10.3/66937$$

$$= 15.385 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 129^2)) 2349 /$$

$$(5/3 + 3 \cdot (17.4) / (8 \cdot 129) - (17.4)^3 / (8 \cdot 129^3))$$

$$= 1356 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 8.43/1356 + (10.3 \cdot 100/669) / 1566$$

$$= 0.016$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-193 B7	
Bolt Allowable Stress	Stba	1400.02	kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.8	mm
User Defined Area in a Single Bolt	UserBoltArea	5.4000	cm ²
BasePlate Allowable Stress	S	1407.13	kgf/cm ²
Area Available in a Single Bolt	BltArea	4.2350	cm ²
Saddle Load Q0 (Weight)	Q0	390.6	kgf
Saddle Load QL (Wind/Seismic contribution)	QL	42.3	kgf
Maximum Transverse Force	Ft	330.2	kgf
Maximum Longitudinal Force	Fl	103.6	kgf
Saddle Bolted to Steel Foundation	No		



Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $104 / (4.23 \cdot 4)$
= 6.1 kgf/cm². Must be less than 1100.0 kgf/cm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $330 / (4.23 \cdot 4)$
= 19.5 kgf/cm². Must be less than 1100.0 kgf/cm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $104 / (1100 \cdot 4)$
= 0.0236 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1200 \cdot 330 + 0$
= 396.29 kgf-m.

Eccentricity (e):
= R_{mom} / QO
= $396 / 391$
= 1014.58 mm > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis
= $1044 / 2 - 50.8$
= 471.20 mm

Modular Ratio Of Steel/Concrete (n1):
= E_S / E_C
= $2074056 / 219500$
= 9.45

K1 = $3(e - 0.5 \cdot B_{plen})$
= $3(1015 - 0.5 \cdot 1044)$
= 1477.73 mm

K2 = $6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$
= $6 \cdot 9.45 \cdot 8.47 / 300 \cdot (471 + 1015)$
= 237819.95 mm²

K3 = $-K2 \cdot (0.5 \cdot B_{plen} + f)$
= $-237820 \cdot (0.5 \cdot 1044 + 471)$
= -236202769.93 mm³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + 1478 \cdot Y^2 + 237820 \cdot Y + -236202768 = 0$
Y = 303.47 mm

Num = $(B_{plen} / 2 - Y / 3 - e)$
= $(1044 / 2 - 303 / 3 - 1015)$
= -593.73

Denom = $(B_{plen} / 2 - Y / 3 + f)$
= $(1044 / 2 - 303 / 3 + 471)$
= 892.04

Total Bolt Tension Force [Tforce]:
= $-QO \cdot \text{Num} / \text{Denom}$
= $-391 \cdot -594 / 892$



= 259.97 kgf

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned} &= Tforce / (Stba \cdot Nbt) \\ &= 260 / (1400 \cdot 2) \\ &= 0.0928 \text{ cm}^2 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0, 0.024, 0.093] \\ &= 0.0928 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$\begin{aligned} &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(391 + 260) / (303 \cdot 300) \\ &= 1.43 \text{ kgf/cm}^2 \end{aligned}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$\begin{aligned} &= (Bplen - WebIngh) / 2 \\ &= (1044 - 993) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

Overturning Moment due To Bolt Tension [Mt]:

$$\begin{aligned} &= Tforce \cdot Adist \\ &= 260 \cdot 2.54 \\ &= 6.60 \text{ kgf-m.} \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned} &= fc \cdot (Y - Adist) / Y \\ &= 1.43 \cdot (303 - 25.4) / 303 \\ &= 1.31 \text{ kgf/cm}^2 \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned} &= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 300 / 6) \cdot (1.31 + 2 \cdot 1.43) \\ &= 1.34 \text{ kgf-m.} \end{aligned}$$

Baseplate Required Thickness [Treq]:

$$\begin{aligned} &= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))} \\ &= \sqrt{(6 \cdot \max(6.6, 1.34) / (300 \cdot 2111))} \\ &= 2.5014 \text{ mm} \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	545.59	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	528.08	cm
Distance from Saddle to Vessel tangent	a1 or a	980.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1200.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		38491.39	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		-51.80	-2114.39
Long. Stress at Top of Midspan		51.80	1407.13
Long. Stress at Bottom of Midspan		35.28	1407.13
Long. Stress at Top of Saddles		162.12	1407.13
Long. Stress at Bottom of Saddles		-102.70	-2114.39
Long. Stress at Bottom of Saddles		102.70	1407.13
Tangential Shear in Shell		149.14	1125.70
Circ. Stress at Horn of Saddle		163.75	1758.91
Circ. Compressive Stress in Shell		13.87	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2001/2 + 0) \cdot 1200/1003$
 $= 3592.3 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(314, 0) \cdot 1200/2940$
 $= 128.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(3956, 0) \cdot 1200/2940$
 $= 1614.6 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (3956/2 + 0) \cdot 1200/1003$
 $= 7099.9 \text{ kgf}$



Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 31391 + \max(128, 3592, 1615, 7100)$$

$$= 38491.4 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1953(0 + \pi/4(1.13 \cdot 1.2)^2)$$

$$= 3079.973 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	38757.37	kgf
Transverse Shear Load	1977.84	kgf
Longitudinal Shear Load	3955.67	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1050/2 + 38 + 16)\sin(120/2), 2 \cdot 546)$$

$$= 1002.857 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{Rm} \cdot L)) / (1 + (4\text{hm}^2)/3L)]$$

$$= -38491 \cdot 98 [1 - (1 - 98/528 + (54.6^2 - 0^2)/(2 \cdot 98 \cdot 528)) / (1 + (4 \cdot 0)/(3 \cdot 528))]$$

$$= -5915.5 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4(1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2)) / (1 + (4\text{hm}^2)/3L) - 4a/L$$

$$= 38491 \cdot 528/4(1 + 2(54.6^2 - 0^2)/(528^2)) / (1 + (4 \cdot 0)/(3 \cdot 528)) - 4 \cdot 98/528$$

$$= 14179.6 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - M2/(\pi \cdot \text{Rm}^2 t)$$

$$= -1.05 \cdot 546/(2 \cdot 34.8) - 14180/(\pi \cdot 546^2 \cdot 34.8)$$

$$= -51.80 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm}/(2t) + M2/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= -1.05 \cdot 546/(2 \cdot 34.8) + 14180/(\pi \cdot 546^2 \cdot 34.8)$$

$$= 35.28 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot \text{Rm}/(2t) - M1/(K1 \cdot \pi \cdot \text{Rm}^2 t)$$

$$= -1.05 \cdot 546/(2 \cdot 34.8) - 5915/(0.11 \cdot \pi \cdot 546^2 \cdot 34.8)$$

$$= 162.12 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot \text{Rm}/(2t) + M1/(K1^* \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= -1.05 \cdot 546/(2 \cdot 34.8) + 5915/(0.19 \cdot \pi \cdot 546^2 \cdot 34.8)$$

$$= -102.70 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot \text{hm}/3))$$

$$= 38491(528 - 2 \cdot 98)/(528 + (4 \cdot 0/3))$$

$$= 24205.1 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:



$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 24205 / (546 \cdot 34.8)$$

$$= 149.14 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(546 \cdot 34.8)}$$

$$= 107.516 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(546 \cdot 34.8)}, 2 \cdot 98)$$

$$= 415.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(1407/1407, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 38491 \cdot 0.1 / (415(34.8 + 1 \cdot 16))$$

$$= -13.87 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.28) [σ7,r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -38491 / (4(34.8 + 1 \cdot 16)415) - 3 \cdot 0.053 \cdot 38491 / (2(34.8 + 1 \cdot 16)^2)$$

$$= -163.75 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1044.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2349.3	kgf
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(528 \cdot 34.8)} + 425)34.8^3) / 12$$

$$= 224.047 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 224 + 222(182 - 17.4)^2$$

$$= 59918.535 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	636.6	34.8	17.4	221.7	386011.3	0.599E+05
Wearplate	425.0	16.0	42.8	68.0	291210.0	0.131E+05
Web	10.0	596.0	348.8	59.6	2078996.5	0.343E+05
BasePlate	300.0	25.0	659.3	75.0	4944937.0	0.171E+06
Totals	...	...	...	424.3	7701155.0	0.279E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7701155/424$$

$$= 181.508 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$



$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 38491$$

$$= 7833.8335 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 38.6666 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1409.5955 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1200 - 528 \cdot \sin(120/2) / 1.05$$

$$= 763.203 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5978.8101 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 38.9508 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1566.2173 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(38491 + 266)300 / (4 \cdot 1044 \cdot 1566)}$$

$$= 23.094 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (525 + 38 + 16)$$

$$= 1002.857 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1003 / (4 - 1)$$

$$= 334.286 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 334 \cdot 300 / 2$$

$$= 501.429 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 38757 / (1044 \cdot 300)$$

$$= 12.375 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 501 \cdot 12.4$$

$$= 6205.000 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19 + 16.7$$

$$= 35.714 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 6205 / 35.7$$

$$= 173.740 \text{ kgf/cm}^2$$

Check of Outside Ribs:



Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 200 - 10
 = 190.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/35.7
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 200/2
 = 100.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{667/35.7}$
 = 43.205 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1200 - cos(120/2)(525 + 38 + 16) - 25
 = 885.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 1984060/2349)}$
 = 129.113

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 886/43.2
 = 20.495

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 3956/ (2 • 1044) • 334 • 886/2
 = 280.393 kgf-m.

Compressive Allowable, KL/r < Cc (20.5 < 129) per AISC E2-1 [Sca]:

= (1-(KLr)²/(2•Cc²))Fy/(5/3+3•(KLr)/(8•Cc)-(KLr)³/(8•Cc³)
 = (1-(20.5)²/(2 • 129²))2349/
 (5/3+3•(20.5)/(8• 129)-(20.5³)/(8•129³)
 = 1344 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 174/1344 + (280 • 100/6666666)/1566
 = 0.398

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	334.3	10.0	0.0	33.4	0.0	2.79
Totals	...	...	...	52.4	...	669.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/52.4
 = 0.000 mm



Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1200 - \sqrt{(579 + 16)^2 - (334/2)^2} - 25$$

$$= 620.650 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{669/52.4}$$

$$= 35.731 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$

$$= 1 \cdot 621/35.7$$

$$= 17.370$$

Unit Force [Force,u]:

$$= \text{FI} / (2 \cdot \text{Baseplate Length})$$

$$= 3956 / (2 \cdot 1044)$$

$$= 1.894 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.89 \cdot 334 \cdot 621$$

$$= 393.056 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 393/66937$$

$$= 587.204 \text{ kgf/cm}^2$$

Compressive Allowable, $\text{KL}/r < C_c$ ($17.4 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot C_c) - (\text{KL}/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.4)^2 / (2 \cdot 129^2)) 2349 /$$

$$(5/3 + 3 \cdot (17.4) / (8 \cdot 129) - (17.4^3) / (8 \cdot 129^3))$$

$$= 1356 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Mbase,c} \cdot C1/I) / \text{Sba}$$

$$= 207/1356 + (393 \cdot 100/669) / 1566$$

$$= 0.528$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	545.59	mm
Shell Thickness used in this Case	t	38.000	mm
Stiffened Vessel Length per 4.15.6	L	528.08	cm
Distance from Saddle to Vessel tangent	a1 or a	980.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	425.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	1407.13	kgf/cm ²
Inside or Mean Depth of Head	Hi or hm	28.21	cm
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	



Distance from Saddle Base to Centerline B 1200.00 mm
 Coefficient of Friction mu 0.30

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case 8449.94 kgf
 Pressure used in Analysis P -1.055 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-15.31	-2114.39
Long. Stress at Top of Midspan	15.31	1407.13
Long. Stress at Bottom of Midspan	-1.21	-2114.39
Long. Stress at Bottom of Midspan	1.21	1407.13
Long. Stress at Top of Saddles	44.22	1407.13
Long. Stress at Bottom of Saddles	-37.35	-2114.39
Long. Stress at Bottom of Saddles	37.35	1407.13
Tangential Shear in Shell	30.56	1125.70
Circ. Stress at Horn of Saddle	35.95	1758.91
Circ. Compressive Stress in Shell	3.05	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (2001/2 + 0) \cdot 1200/1003$
 $= 3592.3 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(314, 0) \cdot 1200/2940$
 $= 128.2 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(3956, 0) \cdot 1200/2940$
 $= 1614.6 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (3956/2 + 0) \cdot 1200/1003$
 $= 7099.9 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 1350 + \max(128, 3592, 1615, 7100)$
 $= 8449.9 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1953 (0 + \pi/4(1.13 \cdot 1.2)^2)$
 $= 3079.973 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 8715.91 kgf
 Transverse Shear Load 1977.84 kgf
 Longitudinal Shear Load 3955.67 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(1050/2 + 38 + 16) \sin(120/2), 2 \cdot 546)$
 $= 1002.857 \text{ mm}$

The Computed K values from Table 4.15.1:



K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)] \\
 &= -8450 \cdot 98 [1 - (1 - 98/528 + (54.6^2 - 28.2^2)/(2 \cdot 98 \cdot 528))/(1 + (4 \cdot 28.2)/(3 \cdot 528))] \\
 &= -1822.3 \text{ kgf-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 (1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/3L) - 4a/L \\
 &= 8450 \cdot 528/4 (1 + 2(54.6^2 - 28.2^2)/(528^2))/(1 + (4 \cdot 28.2)/(3 \cdot 528)) - 4 \cdot 98/528 \\
 &= 2295.8 \text{ kgf-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\
 &= -1.05 \cdot 546/(2 \cdot 34.8) - 2296/(\pi \cdot 546^2 \cdot 34.8) \\
 &= -15.31 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= -1.05 \cdot 546/(2 \cdot 34.8) + 2296/(\pi \cdot 546^2 \cdot 34.8) \\
 &= -1.21 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\
 &= -1.05 \cdot 546/(2 \cdot 34.8) - 1822/(0.11 \cdot \pi \cdot 546^2 \cdot 34.8) \\
 &= 44.22 \text{ kgf/cm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t) \\
 &= -1.05 \cdot 546/(2 \cdot 34.8) + 1822/(0.19 \cdot \pi \cdot 546^2 \cdot 34.8) \\
 &= -37.35 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\
 &= 8450 (528 - 2 \cdot 98)/(528 + (4 \cdot 28.2/3)) \\
 &= 4960.4 \text{ kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (R_m \cdot t) \\
 &= 1.17 \cdot 4960/(546 \cdot 34.8) \\
 &= 30.56 \text{ kgf/cm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\
 &= 0.78 \cdot \sqrt{(546 \cdot 34.8)} \\
 &= 107.516 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\
 &= \min(200 + 1.56 \cdot \sqrt{(546 \cdot 34.8)}, 2 \cdot 98) \\
 &= 415.03 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(S_r/S, 1) \\
 &= \min(1407/1407, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot t_r)) \\
 &= -0.76 \cdot 8450 \cdot 0.1/(415(34.8 + 1 \cdot 16)) \\
 &= -3.05 \text{ kgf/cm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_7, r]$:
 $= -Q/(4(t+\eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q/(2(t+\eta \cdot tr)^2)$
 $= -8450/(4(34.8 + 1 \cdot 16)415) -$
 $3 \cdot 0.053 \cdot 8450/(2(34.8 + 1 \cdot 16)^2)$
 $= -35.95 \text{ kgf/cm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1044.0000	mm
Baseplate Thickness	Bpthk	25.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2349.3	kgf
Height of Web at Center	Hw,c	300.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3)/12$
 $= ((1.56 \cdot \sqrt{(528 \cdot 34.8)} + 425)^3)/12$
 $= 224.047 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 224 + 222(182 - 17.4)^2$
 $= 59918.535 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	636.6	34.8	17.4	221.7	386011.3	0.599E+05
Wearplate	425.0	16.0	42.8	68.0	291210.0	0.131E+05
Web	10.0	596.0	348.8	59.6	2078996.5	0.343E+05
BasePlate	300.0	25.0	659.3	75.0	4944937.0	0.171E+06
Totals	...	...	...	424.3	7701155.0	0.279E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 7701155/424$
 $= 181.508 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2)/(\pi - 2.09 + \sin(120)\cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K_1 \cdot Q$
 $= 0.2 \cdot 8450$
 $= 1719.7455 \text{ kgf}$

Tension Stress, $St = (Fh/As) = 8.4884 \text{ kgf/cm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 1409.5955 \text{ kgf/cm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians})$
 $= 1200 - 528 \cdot \sin(120/2)/1.05$
 $= 763.203 \text{ mm}$

Bending Moment, $M = Fh \cdot d = 1312.5160 \text{ kgf-m.}$



Bending Stress, $S_b = (M \cdot C_1 / I) = 8.5508 \text{ kgf/cm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 1566.2173 \text{ kgf/cm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(8450 + 266) \cdot 300) / (4 \cdot 1044 \cdot 1566)}$
 $= 10.951 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (525 + 38 + 16)$
 $= 1002.857 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1003 / (4 - 1)$
 $= 334.286 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 334 \cdot 300 / 2$
 $= 501.429 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 8716 / (1044 \cdot 300)$
 $= 2.783 \text{ kgf/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 501 \cdot 2.78$
 $= 1395.405 \text{ kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 19 + 16.7$
 $= 35.714 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 1395 / 35.7$
 $= 39.071 \text{ kgf/cm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 200 - 10$
 $= 190.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 35.7$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 200 / 2$
 $= 100.000 \text{ mm}$

Radius of Gyration [r]:



$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{667/35.7}$$

$$= 43.205 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1200 - \cos(120/2) (525 + 38 + 16) - 25$$

$$= 885.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 1984060/2349)}$$

$$= 129.113$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 886/43.2$$

$$= 20.495$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 3956 / (2 \cdot 1044) \cdot 334 \cdot 886/2$$

$$= 280.393 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($20.5 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (20.5)^2 / (2 \cdot 129^2)) 2349 / (5/3 + 3 \cdot (20.5) / (8 \cdot 129) - (20.5)^3 / (8 \cdot 129^3))$$

$$= 1344 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 39.1/1344 + (280 \cdot 100/6666666) / 1566$$

$$= 0.298$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	334.3	10.0	0.0	33.4	0.0	2.79
Totals	...	...	...	52.4	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/52.4$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1200 - \sqrt{(579 + 16)^2 - (334/2)^2} - 25$$

$$= 620.650 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{669/52.4}$$

$$= 35.731 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 621/35.7$$

$$= 17.370$$

Unit Force [Force,u]:



$$\begin{aligned} &= F_l / (2 \cdot \text{Baseplate Length}) \\ &= 3956 / (2 \cdot 1044) \\ &= 1.894 \text{ kgf/mm} \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned} &= \text{Unit Force} \cdot e \cdot L \\ &= 1.89 \cdot 334 \cdot 621 \\ &= 393.056 \text{ kgf-m.} \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned} &= \text{Bending Moment} / \text{Section Modulus} \\ &= 393/66937 \\ &= 587.204 \text{ kgf/cm}^2 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($17.4 < 129$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (17.4)^2 / (2 \cdot 129^2)) 2349 / \\ &\quad (5/3 + 3 \cdot (17.4) / (8 \cdot 129) - (17.4^3) / (8 \cdot 129^3)) \\ &= 1356 \text{ kgf/cm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} &= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba} \\ &= 46.6/1356 + (393 \cdot 100/669) / 1566 \\ &= 0.409 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N-Tube Inlet

From: 20

Pressure for Reinforcement Calculations	P	65.869	kgf/cm ²
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	160	°C

Parent Material [Impact Tested]		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1050.00	mm
Design Length of Section	L	1863.2998	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		75.08	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

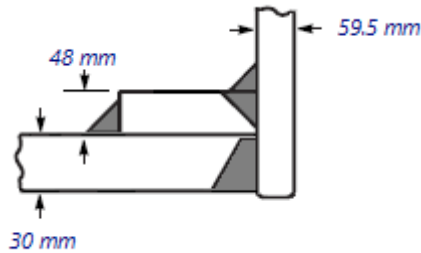
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material [Impact Tested]		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	800.0000	mm
Thickness of Pad	te	48.0000	mm
Weld leg size between Pad and Shell	Wp	36.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	48.0000	mm
Reinforcing Pad Width		95.2000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	490.576 mm.
Actual Thickness Used in Calculation	59.512 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $65.9 \cdot 528 / (1407 \cdot 1 - 0.6 \cdot 65.9)$
= 25.4389 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $65.9 \cdot 248 / (1203 \cdot 1 - 0.6 \cdot 65.9)$
= 14.0640 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.5588 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	993.8511 mm
Parallel to Vessel Wall, opening length	d	496.9256 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		67.0625 mm

Weld Strength Reduction Factor [fr1]:

= min(1, S_n / S_v)
= min(1, 1203/1407)
= 0.855

Weld Strength Reduction Factor [fr2]:

= min(1, S_n / S_v)
= min(1, 1203/1407)
= 0.855

Weld Strength Reduction Factor [fr4]:

= min(1, S_p / S_v)
= min(1, 1407/1407)
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(0.86, 1)
= 0.855

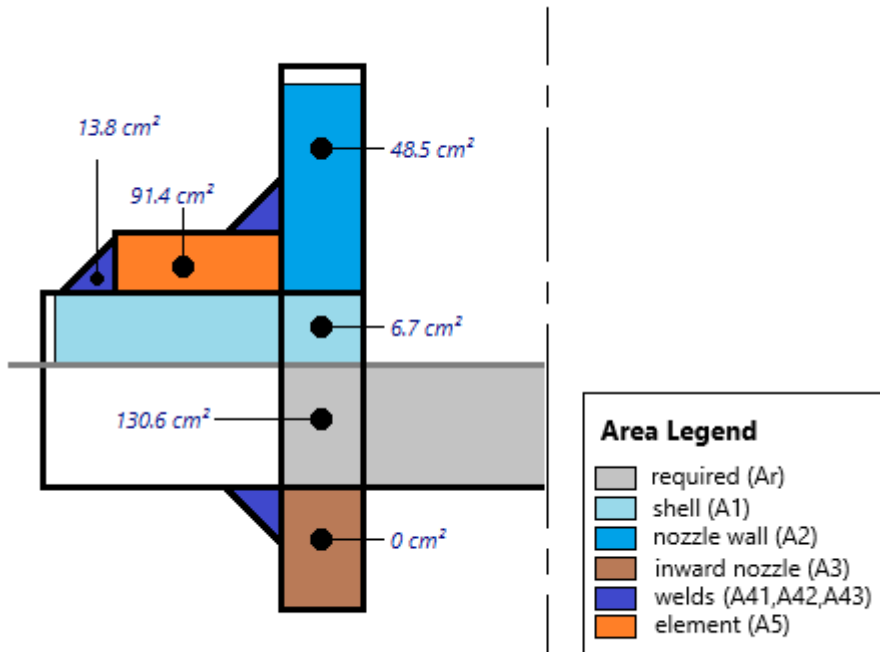


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	130.567	11.389	NA
Area in Shell	A1	6.661	107.592	NA
Area in Nozzle Wall	A2	48.482	62.824	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	13.815	13.815	NA
Area in Element	A5	91.392	91.392	NA
TOTAL AREA AVAILABLE	Atot	160.350	275.623	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 737.9507 48.0000 mm
 Based on given Pad Diameter: 800.0000 32.3573 mm
 Based on Shell or Nozzle Thickness: 814.9611 30.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (497 \cdot 25.4 \cdot 1 + 2 \cdot 56.3 \cdot 25.4 \cdot 1 \cdot (1 - 0.86))$
 $= 130.567 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 497(1 \cdot 26.8 - 1 \cdot 25.4) - 2 \cdot 56.3$
 $(1 \cdot 26.8 - 1 \cdot 25.4) \cdot (1 - 0.86)$
 $= 6.661 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 67.1) \cdot (56.3 - 14.1) \cdot 0.86$



$$= 48.482 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 36^2 \cdot 1$
 $= 13.815 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (800 - 610) \cdot 48 \cdot 1$
 $= 91.392 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$ta = 17.2390 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 5.6750 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 28.6139 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 28.6139 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 3.5710 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 5.6750 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 11.5062 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.5, \max(28.6, 5.67)]$
 $= 11.5062 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(17.2, 11.5)$
 $= 17.2390 \text{ mm}$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 206.4,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2801.6 kgf/cm ²	Passed
Occasional	: 130.5,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 51.8,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $tg = 52.1$, $tr = 14.1$, $c = 3.17 \text{ mm}$, $E* = 1$
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.29$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	37°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $tg = 48$, $c = 3.17 \text{ mm}$, $E* = 1$
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.95$, Temp. Reduction = 3°C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	16°C
Min Metal Temp. at Required thickness (UCS 66.1)	13°C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 30, tr = 25.4, c = 3.17 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.95$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	1 °C

Gov. MDMT of the Nozzle	:	13 °C
Gov. MDMT of the Reinforcement Pad	:	-32 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	13 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $65.87/104.12 = 0.633$

Weld Size Calculations, Description: N-Tube Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$25.4520 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (131 - 6.66 + 2 \cdot 56.3 \cdot 0.86 \cdot (1 \cdot 26.8 - 25.4)) \cdot 1407)$
 $= 176229.47 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (48.5 + 91.4 + 13.8 - 0 \cdot 0.86) \cdot 1407$
 $= 216259.80 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (48.5 + 0 + 0.86 + (25.8)) \cdot 1407$
 $= 105789.89 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (48.5 + 0 + 13.8 + 91.4 + (25.8)) \cdot 1407$
 $= 252626.42 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 1203$
 $= 56455. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 800 \cdot 36 \cdot 0.49 \cdot 1407$
 $= 311919. \text{ kgf}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 277) \cdot (59.5 - 3.17) \cdot 0.7 \cdot 1203$$

$$= 412366. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 610 \cdot 48 \cdot 0.74 \cdot 1407$$

$$= 478598. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 610 \cdot (30 - 3.17) \cdot 0.74 \cdot 1407$$

$$= 267467. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (311919 + 412366) = 724285 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (56455 + 478598 + 267467 + 0) = 802519 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (311919 + 267467 + 0) = 579386 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 724284 kgf, must exceed W = 176229 kgf or W1 = 216259 kgf
 Path 2-2 = 802519 kgf, must exceed W = 176229 kgf or W2 = 105789 kgf
 Path 3-3 = 579385 kgf, must exceed W = 176229 kgf or W3 = 252626 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 69.2 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 97.5401 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 477.5401 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 341.07 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 496.93 mm

Input Echo, WRC297 Item 1, Description: N-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1050.000	mm
Shell Thickness	30.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	490.576	mm
Nozzle Thickness	59.5122	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	



Nozzle Cold S.I. Allowable	SNmc	1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	800.000	mm
Thickness of Reinforcing Pad	48.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	65.87	kgf/cm ²
Radial Load	P	2059.93	kgf
Circumferential Shear	Vc	3059.30	kgf
Longitudinal Shear	Vl	3059.30	kgf
Circumferential Moment	Mc	5504.51	kgf-m.
Longitudinal Moment	Ml	7339.35	kgf-m.
Torsional Moment	Mt	9174.19	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.549) is > 0.5
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{565.588(30.0 - 3.175)}$$

$$= 1016.075 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1131.175	mm
Nozzle Outside Diameter	(d) =	609.600	mm
Vessel Thickness used	(T) =	74.825	mm
Nozzle Thickness used	(t) =	56.337	mm
T / t	=	1.328	
d / t	=	10.821	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.095$$

Nr/P	=	0.100
Mr/P	=	0.118
M0/P	=	0.047
N0/P	=	0.122
MrD/Mc	=	0.234
NrDL/Mc	=	0.155
M0d/Mc	=	0.108
N0DL/Mc	=	0.060
MrD/Ml	=	0.128
NrDL/Ml	=	0.094
M0D/Ml	=	0.065
N0DL/Ml	=	0.144

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-10	10	-10	10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-31	-31	31	31
Outplane Bending (Ml)	-83	83	83	-83



Normal Pressure Stress	434	500	434	500
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Outplane Stress Summary	304	558	534	453
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-26	26	-26	26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-20	-20	20	20
Inplane Bending (ML)	-165	165	165	-165
Inplane Pressure Stress	217	217	217	217

Inplane Stress Summary	1	384	373	94
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	20	20	20	20

Shear Stress Summary	25	25	16	16
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	307	561	535	454
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-10	10	-10	10
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-104	104	104	-104
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	217	217	217	217

Outplane Stress Summary	88	317	316	127
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-26	26	-26	26
Inplane Membrane (Mc)	-24	-24	24	24
Inplane Bending (Mc)	-226	226	226	-226
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	434	500	434	500

Inplane Stress Summary	152	724	656	320
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right



circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4
Torsional Shear (Mt)	20	20	20	20
Shear Stress Summary	16	16	25	25

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	156	724	658	323

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-31	-31	31	31
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	313	392	313	392
Hoop Stress Summary	277	356	339	418

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-26	26	-26	26
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-59	-59	59	59
Axial Bending (ML)	-185	185	185	-185
Axial Pressure Stress	156	156	156	156
Axial Stress Summary	-116	307	371	55

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	5	5	-5	-5
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	27	27	27	27
Shear Stress Summary	33	33	22	22

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	399	373	383	420

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0



Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	313	392	313	392
Hoop Stress Summary	299	378	318	397

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-26	26	-26	26
Axial Membrane (Mc)	-44	-44	44	44
Axial Bending (Mc)	-267	267	267	-267
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	156	156	156	156
Axial Stress Summary	-184	404	439	-42

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	27	27	27	27
Shear Stress Summary	22	22	33	33

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	485	417	448	444

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 609.6 + 2 \cdot 1.65 \cdot \sqrt{541.588(30.0 - 3.175)}$
 $= 1007.357 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	1083.175 mm
Nozzle Outside Diameter (d)	=	800.000 mm
Vessel Thickness used (T)	=	26.825 mm
Nozzle Thickness used (t)	=	151.537 mm
T / t	=	0.177
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.693$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.022
N0/P	=	0.044
MrD/Mc	=	0.247
NrDL/Mc	=	0.089
M0d/Mc	=	0.074
N0DL/Mc	=	0.076
MrD/Ml	=	0.059
NrDL/Ml	=	0.042
M0D/Ml	=	0.018
N0DL/Ml	=	0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-12	-12	-12	-12
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-52	-52	52	52
Outplane Bending (ML)	-138	138	138	-138
Normal Pressure Stress	1264	1330	1264	1330
Outplane Stress Summary	1022	1442	1405	1270

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-12	-12	-12	-12
Inplane Bending (P)	-120	120	-120	120
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-53	-53	53	53
Inplane Bending (ML)	-449	449	449	-449
Inplane Pressure Stress	632	632	632	632
Inplane Stress Summary	-3	1136	1002	344

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	33	33	33	33
Shear Stress Summary	43	43	24	24

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1029	1448	1406	1270

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-12	-12	-12	-12
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	-72	-72	72	72
Outplane Bending (Mc)	-424	424	424	-424
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	632	632	632	632
Outplane Stress Summary	84	1009	1078	306

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-12	-12	-12	-12



Inplane Bending (P)	-120	120	-120	120
Inplane Membrane (Mc)	-84	-84	84	84
Inplane Bending (Mc)	-1419	1419	1419	-1419
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1264	1330	1264	1330

Inplane Stress Summary	-371	2773	2635	104
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	33	33	33	33
Shear Stress Summary	24	24	43	43

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	458	2773	2636	314

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	434	500	434	500	434	500	434	500
Circ. Pl (SUS)	-35	-35	27	27	-27	-27	21	21
Circ. Q (SUS)	-93	93	73	-73	-252	252	200	-200
Long. Pm (SUS)	217	217	217	217	217	217	217	217
Long. Pl (SUS)	-23	-23	17	17	-13	-13	5	5
Long. Q (SUS)	-191	191	139	-139	-114	114	94	-94
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	20	20	20	20	20	20	20	20
Pm (SUS)	434.0	500.0	434.0	500.0	434.0	500.0	434.0	500.0
Pm+Pl (SUS)	399.1	465.1	461.1	527.1	407.1	473.1	455.1	521.1
Pm+Pl+Q (Total)	307.9	561.3	535.6	454.7	158.7	725.6	656.7	323.9

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	500.00	1407.13	Passed
Pm+Pl (SUS)	527.05	2110.69	Passed
Pm+Pl+Q (TOTAL)	725.63	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of	Stress Values at
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Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1264	1330	1264	1330	1264	1330	1264	1330
Circ. Pl (SUS)	-64	-64	40	40	-96	-96	72	72
Circ. Q (SUS)	-176	176	100	-100	-1539	1539	1299	-1299
Long. Pm (SUS)	632	632	632	632	632	632	632	632
Long. Pl (SUS)	-65	-65	41	41	-84	-84	60	60
Long. Q (SUS)	-569	569	329	-329	-462	462	386	-386
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	33	33	33	33	33	33	33	33
Pm (SUS)	1264	1330	1264	1330	1264	1330	1264	1330
Pm+Pl (SUS)	1200	1266	1304	1370	1168	1234	1336	1402
Pm+Pl+Q (Total)	1029	1447	1405	1270	459	2773	2636	314

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1330.00	1407.13	Passed
Pm+Pl (SUS)	1402.11	2110.69	Passed
Pm+Pl+Q (TOTAL)	2773.33	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	313	392	313	392	313	392	313	392
Circ. Pl (SUS)	-35	-35	27	27	-13	-13	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	156	156	156	156	156	156	156	156
Long. Pl (SUS)	-61	-61	57	57	-46	-46	42	42
Long. Q (SUS)	-211	211	159	-159	-293	293	241	-241
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	27	27	27	27	27	27	27	27
Pm (SUS)	313.0	392.0	313.0	392.0	313.0	392.0	313.0	392.0
Pm+Pl (SUS)	278.1	357.1	340.2	419.1	300.1	379.1	318.2	397.1
Pm+Pl+Q (Total)	399.2	372.4	383.2	420.3	485.0	416.1	446.9	444.6

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	392.00	1203.20	Passed
Pm+Pl (SUS)	419.12	1804.80	Passed
Pm+Pl+Q (TOTAL)	485.00	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 27 Licensee: SL User

FileName : E-30003-AK

Nozzle Calcs.: N-Tube Inlet

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Nozl: 9 4:49pm May 21,2026



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Input, Nozzle Desc: N2-Tube Outlet

From: 20

Pressure for Reinforcement Calculations	P	65.765	kgf/cm ²
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	160	°C

Parent Material [Impact Tested]		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1050.00	mm
Design Length of Section	L	1863.2998	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		75.08	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

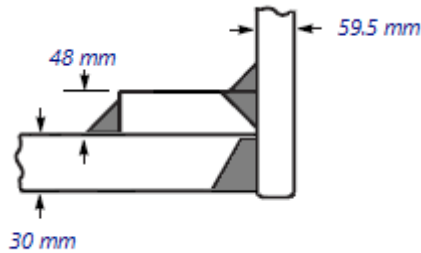
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material [Impact Tested]		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	800.0000	mm
Thickness of Pad	te	48.0000	mm
Weld leg size between Pad and Shell	Wp	36.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	48.0000	mm
Reinforcing Pad Width		95.2000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	490.576 mm.
Actual Thickness Used in Calculation	59.512 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 65.8 \cdot 528 / (1407 \cdot 1 - 0.6 \cdot 65.8)$
 $= 25.3975 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 65.8 \cdot 248 / (1203 \cdot 1 - 0.6 \cdot 65.8)$
 $= 14.0410 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.5588 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	993.8511 mm
Parallel to Vessel Wall, opening length	d	496.9256 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		67.0625 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 1407 / 1407)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

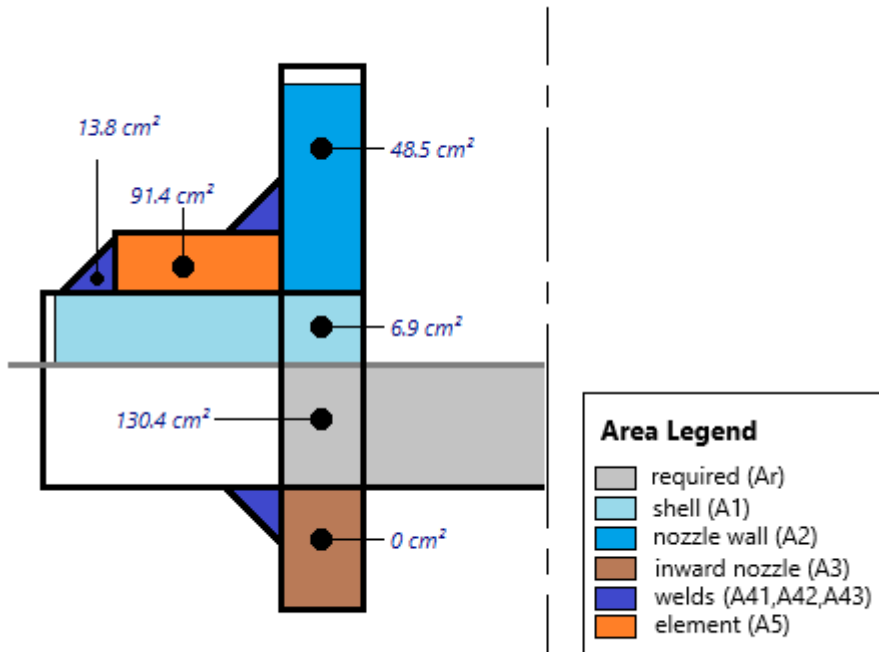


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	130.354	11.389	NA
Area in Shell	A1	6.860	107.592	NA
Area in Nozzle Wall	A2	48.508	62.824	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	13.815	13.815	NA
Area in Element	A5	91.392	91.392	NA
TOTAL AREA AVAILABLE	Atot	160.576	275.623	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 737.0382 48.0000 mm
 Based on given Pad Diameter: 800.0000 32.1273 mm
 Based on Shell or Nozzle Thickness: 813.5011 30.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (497 \cdot 25.4 \cdot 1 + 2 \cdot 56.3 \cdot 25.4 \cdot 1 \cdot (1 - 0.86))$
 $= 130.354 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 497(1 \cdot 26.8 - 1 \cdot 25.4) - 2 \cdot 56.3$
 $(1 \cdot 26.8 - 1 \cdot 25.4) \cdot (1 - 0.86)$
 $= 6.860 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 67.1) \cdot (56.3 - 14) \cdot 0.86$



$$= 48.508 \text{ cm}^2$$

$$\begin{aligned} \text{Area Available in Welds [A41 + A42 + A43]:} \\ &= W_o^2 \cdot fr_3 + (W_i - c_n / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4 \\ &= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 36^2 \cdot 1 \\ &= 13.815 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area Available in Element [A5]:} \\ &= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot fr_4 \\ &= (800 - 610) \cdot 48 \cdot 1 \\ &= 91.392 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} \quad t_a &= 17.2160 \text{ mm} \\ \text{Wall Thickness per UG16(b),} \quad t_{r16b} &= 5.6750 \text{ mm} \\ \text{Wall Thickness, shell/head, internal pressure} \quad t_{rb1} &= 28.5725 \text{ mm} \\ \text{Wall Thickness} \quad t_{b1} &= \max(t_{rb1}, t_{r16b}) = 28.5725 \text{ mm} \\ \text{Wall Thickness, shell/head, external pressure} \quad t_{rb2} &= 3.5710 \text{ mm} \\ \text{Wall Thickness} \quad t_{b2} &= \max(t_{rb2}, t_{r16b}) = 5.6750 \text{ mm} \\ \text{Wall Thickness per table UG-45} \quad t_{b3} &= 11.5062 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Determine Nozzle Thickness candidate [tb]:} \\ &= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ &= \min[11.5, \max(28.6, 5.67)] \\ &= 11.5062 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Minimum Wall Thickness of Nozzle Necks [tUG-45]:} \\ &= \max(t_a, t_b) \\ &= \max(17.2, 11.5) \\ &= 17.2160 \text{ mm} \end{aligned}$$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 206.2,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2801.8 kgf/cm ²	Passed
Occasional	: 130.3,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 51.8,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\begin{aligned} \text{Govrn. thk, } t_g &= 52.1, t_r = 14, c = 3.17 \text{ mm, } E^* = 1 \\ \text{Thickness Ratio} &= t_r \cdot E^* / (t_g - c) = 0.29, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C} \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\begin{aligned} \text{Govrn. thk, } t_g &= 48, c = 3.17 \text{ mm, } E^* = 1 \\ \text{Thickness Ratio} &= t_r \cdot E^* / (t_g - c) = 0.95, \text{Temp. Reduction} = 3 \text{ }^\circ\text{C} \\ \text{Attachment governing, Attachment stress} &= \text{Shell stress per Code} \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	13 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 30, tr = 25.4, c = 3.17 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.95$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	1 °C

Gov. MDMT of the Nozzle	:	13 °C
Gov. MDMT of the Reinforcement Pad	:	-32 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	13 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $65.76/104.12 = 0.632$

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	19.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$25.4520 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (130 - 6.86 + 2 \cdot 56.3 \cdot 0.86 \cdot (1 \cdot 26.8 - 25.4)) \cdot 1407)$
 $= 175706.45 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (48.5 + 91.4 + 13.8 - 0 \cdot 0.86) \cdot 1407$
 $= 216296.94 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (48.5 + 0 + 0.86 + (25.8)) \cdot 1407$
 $= 105827.02 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (48.5 + 0 + 13.8 + 91.4 + (25.8)) \cdot 1407$
 $= 252663.56 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 1203$
 $= 56455. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 800 \cdot 36 \cdot 0.49 \cdot 1407$
 $= 311919. \text{ kgf}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 277) \cdot (59.5 - 3.17) \cdot 0.7 \cdot 1203$$

$$= 412366. \text{ kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 610 \cdot 48 \cdot 0.74 \cdot 1407$$

$$= 478598. \text{ kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 610 \cdot (30 - 3.17) \cdot 0.74 \cdot 1407$$

$$= 267467. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (311919 + 412366) = 724285 \text{ kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (56455 + 478598 + 267467 + 0) = 802519 \text{ kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (311919 + 267467 + 0) = 579386 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 724284 kgf, must exceed W = 175706 kgf or W1 = 216296 kgf
 Path 2-2 = 802519 kgf, must exceed W = 175706 kgf or W2 = 105827 kgf
 Path 3-3 = 579385 kgf, must exceed W = 175706 kgf or W3 = 252663 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 69.2 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 97.5401 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 477.5401 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1050.000	mm
Shell Thickness	30.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1407.13 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1407.13 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	490.576	mm
Nozzle Thickness	59.5122	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	1203.20 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1203.20 kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter of Reinforcing Pad 800.000 mm
 Thickness of Reinforcing Pad 48.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	65.76	kgf/cm ²
Radial Load	P	2059.93	kgf
Circumferential Shear	Vc	3059.30	kgf
Longitudinal Shear	Vl	3059.30	kgf
Circumferential Moment	Mc	5504.51	kgf-m.
Longitudinal Moment	Ml	7339.35	kgf-m.
Torsional Moment	Mt	9174.19	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.549) is > 0.5
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 609.6 + 2 \cdot 1.65 \cdot \sqrt{565.588(30.0 - 3.175)}$
 $= 1016.075 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1131.175 mm
Nozzle Outside Diameter	(d) =	609.600 mm
Vessel Thickness used	(T) =	74.825 mm
Nozzle Thickness used	(t) =	56.337 mm
T / t	=	1.328
d / t	=	10.821

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.095$$

Nr/P	=	0.100
Mr/P	=	0.118
M0/P	=	0.047
N0/P	=	0.122
MrD/Mc	=	0.234
NrDL/Mc	=	0.155
M0d/Mc	=	0.108
N0DL/Mc	=	0.060
MrD/Ml	=	0.128
NrDL/Ml	=	0.094
M0D/Ml	=	0.065
N0DL/Ml	=	0.144

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-10	10	-10	10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-31	-31	31	31
Outplane Bending (ML)	-83	83	83	-83
Normal Pressure Stress	433	499	433	499
Outplane Stress Summary	304	557	533	452



Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-26	26	-26	26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-20	-20	20	20
Inplane Bending (ML)	-165	165	165	-165
Inplane Pressure Stress	216	216	216	216
Inplane Stress Summary	1	384	372	93

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	4	4	-4	-4
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	20	20	20	20
Shear Stress Summary	25	25	16	16

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	307	561	535	453

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-10	10	-10	10
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-104	104	104	-104
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	216	216	216	216
Outplane Stress Summary	87	317	316	127

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-26	26	-26	26
Inplane Membrane (Mc)	-24	-24	24	24
Inplane Bending (Mc)	-226	226	226	-226
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	433	499	433	499
Inplane Stress Summary	152	723	655	319

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-4	-4	4	4



Torsional Shear (Mt)	20	20	20	20
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Shear Stress Summary	16	16	25	25
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	156	724	657	323

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-31	-31	31	31
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	312	391	312	391
Hoop Stress Summary	277	356	339	418

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-26	26	-26	26
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-59	-59	59	59
Axial Bending (ML)	-185	185	185	-185
Axial Pressure Stress	156	156	156	156
Axial Stress Summary	-116	306	371	54

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	5	5	-5	-5
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	27	27	27	27
Shear Stress Summary	33	33	22	22

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	399	372	382	419

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-4	-4	-4	-4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	312	391	312	391



Hoop Stress Summary 298 377 317 396

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-2	-2	-2	-2
Axial Bending (P)	-26	26	-26	26
Axial Membrane (Mc)	-44	-44	44	44
Axial Bending (Mc)	-267	267	267	-267
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	156	156	156	156
Axial Stress Summary	-184	404	439	-42

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-5	-5	5	5
Shear due to Torsion	27	27	27	27
Shear Stress Summary	22	22	33	33

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	485	417	448	444

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 609.6 + 2 \cdot 1.65 \cdot \sqrt{541.588(30.0 - 3.175)}$
 $= 1007.357 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1083.175 mm
 Nozzle Outside Diameter (d) = 800.000 mm
 Vessel Thickness used (T) = 26.825 mm
 Nozzle Thickness used (t) = 151.537 mm
 T / t = 0.177
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.693$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.044
 Mr/P = 0.070
 M0/P = 0.022
 N0/P = 0.044
 MrD/Mc = 0.247
 NrDL/Mc = 0.089
 M0d/Mc = 0.074
 N0DL/Mc = 0.076
 MrD/Ml = 0.059
 NrDL/Ml = 0.042
 M0D/Ml = 0.018
 N0DL/Ml = 0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stresses Normal to	Au Top	Al Top	Bu Bottom	Bl Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	-12	-12	-12	-12
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-52	-52	52	52
Outplane Bending (ML)	-138	138	138	-138
Normal Pressure Stress	1262	1328	1262	1328
Outplane Stress Summary	1020	1440	1403	1268

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-12	-12	-12	-12
Inplane Bending (P)	-120	120	-120	120
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-53	-53	53	53
Inplane Bending (ML)	-449	449	449	-449
Inplane Pressure Stress	631	631	631	631
Inplane Stress Summary	-4	1135	1001	343

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	33	33	33	33
Shear Stress Summary	43	43	24	24

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1028	1445	1404	1268

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-12	-12	-12	-12
Outplane Bending (P)	-38	38	-38	38
Outplane Membrane (Mc)	-72	-72	72	72
Outplane Bending (Mc)	-424	424	424	-424
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	631	631	631	631
Outplane Stress Summary	83	1008	1077	305

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	-12	-12	-12	-12
Inplane Bending (P)	-120	120	-120	120
Inplane Membrane (Mc)	-84	-84	84	84
Inplane Bending (Mc)	-1419	1419	1419	-1419
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1262	1328	1262	1328
Inplane Stress Summary	-373	2771	2633	102

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	33	33	33	33
Shear Stress Summary	24	24	43	43

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	459	2771	2634	313

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	433	499	433	499	433	499	433	499
Circ. Pl (SUS)	-35	-35	27	27	-27	-27	21	21
Circ. Q (SUS)	-93	93	73	-73	-252	252	200	-200
Long. Pm (SUS)	216	216	216	216	216	216	216	216
Long. Pl (SUS)	-23	-23	17	17	-13	-13	5	5
Long. Q (SUS)	-191	191	139	-139	-114	114	94	-94
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	20	20	20	20	20	20	20	20
Pm (SUS)	433.0	499.0	433.0	499.0	433.0	499.0	433.0	499.0
Pm+Pl (SUS)	398.1	464.1	460.1	526.1	406.1	472.1	454.1	520.1
Pm+Pl+Q (Total)	306.9	560.3	534.6	453.7	157.7	724.6	655.7	322.9

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	499.00	1407.13	Passed
Pm+Pl (SUS)	526.05	2110.69	Passed
Pm+Pl+Q (TOTAL)	724.63	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1262	1328	1262	1328	1262	1328	1262	1328



Circ. P1 (SUS)	-64	-64	40	40	-96	-96	72	72
Circ. Q (SUS)	-176	176	100	-100	-1539	1539	1299	-1299
Long. Pm (SUS)	631	631	631	631	631	631	631	631
Long. P1 (SUS)	-65	-65	41	41	-84	-84	60	60
Long. Q (SUS)	-569	569	329	-329	-462	462	386	-386
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	33	33	33	33	33	33	33	33
Pm (SUS)	1262	1328	1262	1328	1262	1328	1262	1328
Pm+P1 (SUS)	1198	1264	1302	1368	1166	1232	1334	1400
Pm+P1+Q (Total)	1028	1445	1403	1268	460	2771	2634	313

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1328.00	1407.13	Passed
Pm+P1 (SUS)	1400.11	2110.69	Passed
Pm+P1+Q (TOTAL)	2771.33	4221.39	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	312	391	312	391	312	391	312	391
Circ. P1 (SUS)	-35	-35	27	27	-13	-13	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	156	156	156	156	156	156	156	156
Long. P1 (SUS)	-61	-61	57	57	-46	-46	42	42
Long. Q (SUS)	-211	211	159	-159	-293	293	241	-241
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	27	27	27	27	27	27	27	27
Pm (SUS)	312.0	391.0	312.0	391.0	312.0	391.0	312.0	391.0
Pm+P1 (SUS)	277.1	356.1	339.2	418.1	299.1	378.1	317.2	396.1
Pm+P1+Q (Total)	398.2	371.6	383.0	419.3	484.0	415.8	446.9	443.6

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	391.00	1203.20	Passed
Pm+P1 (SUS)	418.12	1804.80	Passed
Pm+P1+Q (TOTAL)	484.00	3609.61	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: N3-Shell Inlet

From: 50

Pressure for Reinforcement Calculations	P	85.599	kgf/cm ²
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	160	°C

Parent Material [Impact Tested]		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1050.00	mm
Design Length of Section	L	5368.3003	mm
Shell Finished (Minimum) Thickness	t	38.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		283.03	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		300.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
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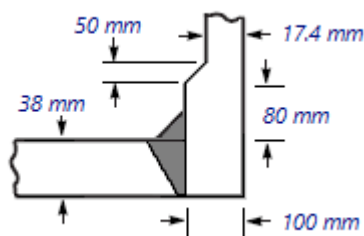
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	50.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm

Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: N3-Shell Inlet**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 288.950 mm.
 Actual Thickness Used in Calculation 17.450 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 85.6 \cdot 528 / (1407 \cdot 1 - 0.6 \cdot 85.6)$
 $= 33.3474$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 85.6 \cdot 148 / (1203 \cdot 1 - 0.6 \cdot 85.6)$
 $= 10.9726$ mm

Required Nozzle thickness under External Pressure per UG-28 : 1.0073 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 82.6 / 0.5773$
 $= 142.9812$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

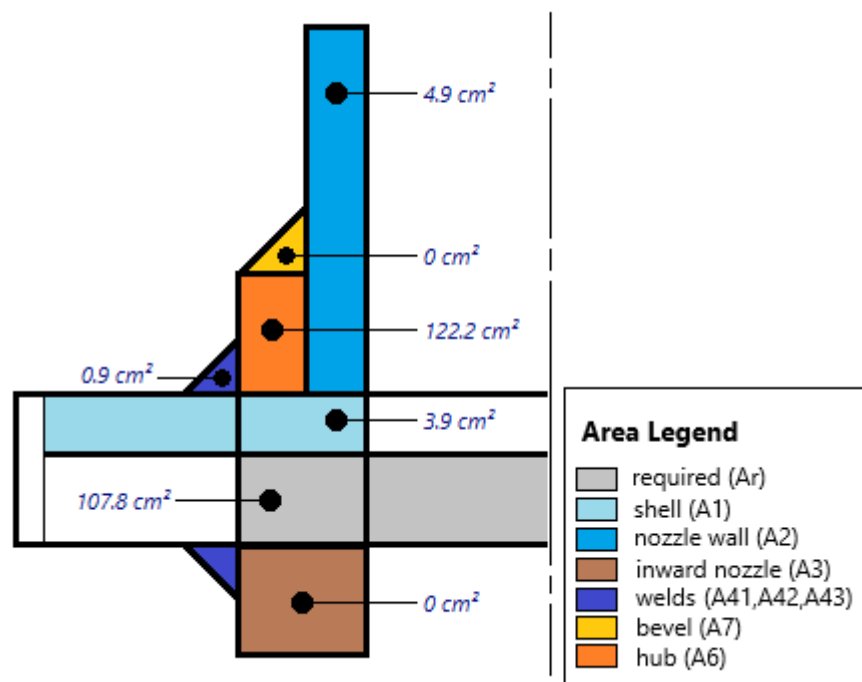
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	590.6008	mm
Parallel to Vessel Wall, opening length	d	295.3004	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	87.0625	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	107.834	11.274	NA
Area in Shell	A1	3.949	74.431	NA
Area in Nozzle Wall	A2	4.917	19.754	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		0.855	0.855	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	122.205	122.205	NA
TOTAL AREA AVAILABLE	Atot	131.925	217.245	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (295 \cdot 33.3 \cdot 1 + 2 \cdot 96.8 \cdot 33.3 \cdot 1 \cdot (1 - 0.86))$
 $= 107.834 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 295 (1 \cdot 34.8 - 1 \cdot 33.3) - 2 \cdot 96.8$
 $(1 \cdot 34.8 - 1 \cdot 33.3) \cdot (1 - 0.86)$
 $= 3.949 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 87.1) (14.3 - 11) 0.86$
 $= 4.917 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 0.855 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(87.1, 300, 80)) \cdot (100 - 17.4) \cdot 0.86 \\ &= 122.205 \text{ cm}^2 \text{ Includes Bevel Area of } 9.266 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\begin{aligned} \text{Wall Thickness for Internal/External pressures} & \quad t_a = 14.1476 \text{ mm} \\ \text{Wall Thickness per UG16(b),} & \quad t_{r16b} = 5.6750 \text{ mm} \\ \text{Wall Thickness, shell/head, internal pressure} & \quad t_{rb1} = 36.5224 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b1} = \max(t_{rb1}, t_{r16b}) = 36.5224 \text{ mm} \\ \text{Wall Thickness, shell/head, external pressure} & \quad t_{rb2} = 3.5710 \text{ mm} \\ \text{Wall Thickness} & \quad t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.6750 \text{ mm} \\ \text{Wall Thickness per table UG-45} & \quad t_{b3} = 11.5062 \text{ mm} \end{aligned}$$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ &= \min[11.5, \max(36.5, 5.67)] \\ &= 11.5062 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(14.1, 11.5) \\ &= 14.1476 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 17.450 = 15.269 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 652.5,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2355.5 kgf/cm ²	Passed
Occasional	: 422.3,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 150.0,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 15.3$, $t_r = 11$, $c = 3.17 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.91$, Temp. Reduction = 5°C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-1 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 38$, $t_r = 33.3$, $c = 3.17 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.96$, Temp. Reduction = 2°C

Min Metal Temp. w/o impact per UCS-66, Curve B	10 °C
Min Metal Temp. at Required thickness (UCS 66.1)	8 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 8 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-28 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $85.60 / 104.12 = 0.822$

Weld Size Calculations, Description: N3-Shell Inlet

Intermediate Calc. for nozzle/shell welds $T_{min} \quad 19.0000 \text{ mm}$

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (108 - 3.95 + 2 \cdot 96.8 \cdot 0.86 \cdot (1 \cdot 34.8 - 33.3)) \cdot 1407)$
 $= 149622.31 \text{ kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (127 + 0.86 - 0 \cdot 0.86) \cdot 1407$
 $= 180079.28 \text{ kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (127 + 0 + 0.86 + (57.7)) \cdot 1407$
 $= 261221.42 \text{ kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (127 + 0 + 0.86 + 0 + (57.7)) \cdot 1407$
 $= 261221.42 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 489 \cdot 10 \cdot 0.49 \cdot 1203$
 $= 45281. \text{ kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 196) \cdot (100 - 3.17) \cdot 0.7 \cdot 1203$
 $= 502306. \text{ kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 489 \cdot (38 - 3.17) \cdot 0.74 \cdot 1407$
 $= 278510. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (45281 + 502306) = 547587 \text{ kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (45281 + 0 + 278510 + 0) = 323791 \text{ kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (45281 + 278510 + 0) = 323791 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 547587 kgf, must exceed W = 149622 kgf	or W1 = 180079 kgf
Path 2-2 = 323791 kgf, must exceed W = 149622 kgf	or W2 = 261221 kgf
Path 3-3 = 323791 kgf, must exceed W = 149622 kgf	or W3 = 261221 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 89.1 kgf/cm²



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 60.3960 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 398.3960 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1050.000	mm
Shell Thickness	38.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	288.950	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	85.60	kgf/cm ²
Radial Load	P	1029.96	kgf
Circumferential Shear	Vc	1529.65	kgf
Longitudinal Shear	Vl	1529.65	kgf
Circumferential Moment	Mc	1376.13	kgf-m.
Longitudinal Moment	Ml	1834.84	kgf-m.
Torsional Moment	Mt	2293.54	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.890) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 488.95 + 2 \cdot 1.65 \cdot \sqrt{(545.588(38.0 - 3.175))}$
 $= 943.825 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1091.175	mm
Nozzle Outside Diameter	(d) =	488.950	mm
Vessel Thickness used	(T) =	34.825	mm
Nozzle Thickness used	(t) =	96.825	mm
T / t	=	0.360	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.508$$



Nr/P = 0.063
 Mr/P = 0.121
 M0/P = 0.037
 N0/P = 0.080
 MrD/Mc = 0.276
 NrDL/Mc = 0.097
 M0d/Mc = 0.084
 N0DL/Mc = 0.081
 MrD/Ml = 0.099
 NrDL/Ml = 0.065
 M0D/Ml = 0.029
 N0DL/Ml = 0.100

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-18	18	-18	18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-30	-30	30	30
Outplane Bending (ML)	-53	53	53	-53
Normal Pressure Stress	1256	1342	1256	1342
Outplane Stress Summary	1147	1377	1315	1331

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-61	61	-61	61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-20	-20	20	20
Inplane Bending (ML)	-183	183	183	-183
Inplane Pressure Stress	628	628	628	628
Inplane Stress Summary	358	847	764	521

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	17	17	17	17
Shear Stress Summary	23	23	11	11

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1147	1378	1315	1331

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-18	18	-18	18
Outplane Membrane (Mc)	-18	-18	18	18



Outplane Bending (Mc)	-116	116	116	-116
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	628	628	628	628

Outplane Stress Summary	467	738	738	542
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-61	61	-61	61
Inplane Membrane (Mc)	-22	-22	22	22
Inplane Bending (Mc)	-383	383	383	-383
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1256	1342	1256	1342

Inplane Stress Summary	783	1759	1595	1037
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	17	17	17	17

Shear Stress Summary	11	11	23	23
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	784	1759	1596	1038
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-30	-30	30	30
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	117	220	117	220

Hoop Stress Summary	80	183	141	244
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-5	5	-5	5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-15	15	15	-15
Axial Pressure Stress	58	58	58	58

Axial Stress Summary	17	61	86	66
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	8	8	4	4

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	81	183	142	244

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	117	220	117	220
Hoop Stress Summary	92	195	129	232

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-5	5	-5	5
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-40	40	40	-40
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	58	58	58	58
Axial Stress Summary	-2	91	106	36

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	8	8

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	95	195	132	232

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	1256	1342	1256	1342	1256	1342	1256	1342
Circ. Pl (SUS)	-36	-36	24	24	-27	-27	17	17
Circ. Q (SUS)	-71	71	35	-35	-444	444	322	-322
Long. Pm (SUS)	628	628	628	628	628	628	628	628
Long. Pl (SUS)	-25	-25	15	15	-24	-24	12	12
Long. Q (SUS)	-244	244	122	-122	-134	134	98	-98
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	1256	1342	1256	1342	1256	1342	1256	1342
Pm+Pl (SUS)	1220	1306	1280	1366	1229	1315	1273	1359
Pm+Pl+Q (Total)	1149	1377	1315	1331	785	1759	1595	1037

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1342.00	1407.13	Passed
Pm+Pl (SUS)	1366.03	2110.69	Passed
Pm+Pl+Q (TOTAL)	1759.14	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	117	220	117	220	117	220	117	220
Circ. Pl (SUS)	-36	-36	24	24	-24	-24	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-18	-18	18	18	-13	-13	13	13
Long. Q (SUS)	-20	20	10	-10	-45	45	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	117.0	220.0	117.0	220.0	117.0	220.0	117.0	220.0
Pm+Pl (SUS)	81.1	184.0	141.1	244.0	93.1	196.0	129.1	232.0
Pm+Pl+Q (Total)	82.0	184.5	141.3	244.1	93.3	196.2	131.5	232.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	220.00	1203.20	Passed
Pm+Pl (SUS)	244.02	1804.80	Passed
Pm+Pl+Q (TOTAL)	244.09	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: N4-Shell Outlet

From: 50

Pressure for Reinforcement Calculations	P	85.495	kgf/cm ²
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	160	°C

Parent Material [Impact Tested]		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	1050.00	mm
Design Length of Section	L	5368.3003	mm
Shell Finished (Minimum) Thickness	t	38.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 633.03 cm

User Entered Minimum Design Metal Temperature -28.89 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²

Diameter Basis (for tr calc only)	Inside		
Layout Angle	90.00	deg	
Diameter	300.0000	mm	

Size and Thickness Basis	Nominal		
Nominal Thickness	80		

Flange Material SA-105

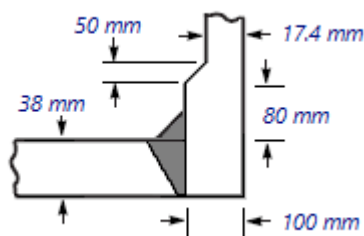
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	50.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm

Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	38.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class	600		
Flange Grade	GR 1.1		

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: N4-Shell Outlet**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 288.950 mm.
 Actual Thickness Used in Calculation 17.450 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 85.5 \cdot 528 / (1407 \cdot 1 - 0.6 \cdot 85.5)$
 $= 33.3052 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 85.5 \cdot 148 / (1203 \cdot 1 - 0.6 \cdot 85.5)$
 $= 10.9587 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0073 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 82.6 / 0.5773$
 $= 142.9812 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

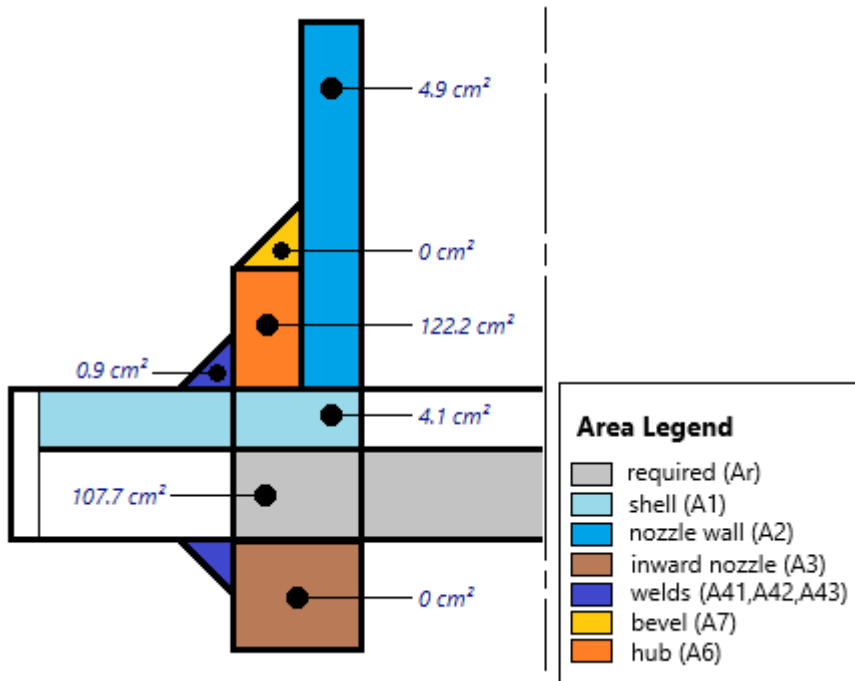
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	590.6008	mm
Parallel to Vessel Wall, opening length	d	295.3004	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	87.0625	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203 / 1407)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	107.698	11.274	NA
Area in Shell	A1	4.061	74.431	NA
Area in Nozzle Wall	A2	4.937	19.754	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		0.855	0.855	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	122.205	122.205	NA
TOTAL AREA AVAILABLE	Atot	132.059	217.245	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (295 \cdot 33.3 \cdot 1 + 2 \cdot 96.8 \cdot 33.3 \cdot 1 \cdot (1 - 0.86))$
 $= 107.698 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 295(1 \cdot 34.8 - 1 \cdot 33.3) - 2 \cdot 96.8$
 $(1 \cdot 34.8 - 1 \cdot 33.3) \cdot (1 - 0.86)$
 $= 4.061 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 87.1)(14.3 - 11)0.86$
 $= 4.937 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 0.855 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(87.1, 300, 80)) \cdot (100 - 17.4) \cdot 0.86$$

$$= 122.205 \text{ cm}^2 \text{ Includes Bevel Area of } 9.266 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 14.1337 mm
Wall Thickness per UG16(b),	tr16b = 5.6750 mm
Wall Thickness, shell/head, internal pressure	trb1 = 36.4802 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 36.4802 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.5710 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.6750 mm
Wall Thickness per table UG-45	tb3 = 11.5062 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.5, \max(36.5, 5.67)]$$

$$= 11.5062 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(14.1, 11.5)$$

$$= 14.1337 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 17.450 = 15.269 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 652.0,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2356.0 kgf/cm ²	Passed
Occasional	: 421.8,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 150.0,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 15.3, tr = 11, c = 3.17 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.91$, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-1 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 38, tr = 33.3, c = 3.17 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.96$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	10 °C
Min Metal Temp. at Required thickness (UCS 66.1)	8 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 8 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-28 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $85.49/104.12 = 0.821$

Weld Size Calculations, Description: N4-Shell Outlet

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (108 - 4.06 + 2 • 96.8 • 0.86 •
(1 • 34.8 - 33.3))1407)
= 149370.33 kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (127 + 0.86 - 0 • 0.86) • 1407
= 180108.52 kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (127 + 0 + 0.86 + (57.7)) • 1407
= 261250.67 kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (127 + 0 + 0.86 + 0 + (57.7)) • 1407
= 261250.67 kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 489 • 10 • 0.49 • 1203
= 45281. kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 196) • (100 - 3.17) • 0.7 • 1203
= 502306. kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 489 • (38 - 3.17) • 0.74 • 1407
= 278510. kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (45281 + 502306) = 547587 kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (45281 + 0 + 278510 + 0) = 323791 kgf

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (45281 + 278510 + 0) = 323791 kgf

Summary of Failure Path Calculations:

Path 1-1 = 547587 kgf, must exceed W = 149370 kgf	or W1 = 180108 kgf
Path 2-2 = 323791 kgf, must exceed W = 149370 kgf	or W2 = 261250 kgf
Path 3-3 = 323791 kgf, must exceed W = 149370 kgf	or W3 = 261250 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 89.1 kgf/cm²



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1.05 kgf/cm²

The Drop for this Nozzle is : 60.3960 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 398.3960 mm

Input Echo, WRC297 Item 1, Description: N4-Shell Outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1050.000	mm
Shell Thickness	38.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	288.950	mm
Nozzle Thickness	100.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	85.49	kgf/cm ²
Radial Load	P	1029.96	kgf
Circumferential Shear	Vc	1529.65	kgf
Longitudinal Shear	Vl	1529.65	kgf
Circumferential Moment	Mc	1376.13	kgf-m.
Longitudinal Moment	Ml	1834.84	kgf-m.
Torsional Moment	Mt	2293.54	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.890) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 488.95 + 2 \cdot 1.65 \cdot \sqrt{(545.588(38.0 - 3.175))}$$

$$= 943.825 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1091.175	mm
Nozzle Outside Diameter	(d) =	488.950	mm
Vessel Thickness used	(T) =	34.825	mm
Nozzle Thickness used	(t) =	96.825	mm
T / t	=	0.360	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.508$$



Nr/P = 0.063
 Mr/P = 0.121
 M0/P = 0.037
 N0/P = 0.080
 MrD/Mc = 0.276
 NrDL/Mc = 0.097
 M0d/Mc = 0.084
 N0DL/Mc = 0.081
 MrD/Ml = 0.099
 NrDL/Ml = 0.065
 M0D/Ml = 0.029
 N0DL/Ml = 0.100

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-18	18	-18	18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-30	-30	30	30
Outplane Bending (ML)	-53	53	53	-53
Normal Pressure Stress	1255	1340	1255	1340
Outplane Stress Summary	1145	1375	1313	1330

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-61	61	-61	61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-20	-20	20	20
Inplane Bending (ML)	-183	183	183	-183
Inplane Pressure Stress	627	627	627	627
Inplane Stress Summary	357	847	763	520

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	17	17	17	17
Shear Stress Summary	23	23	11	11

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1146	1376	1314	1330

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-18	18	-18	18
Outplane Membrane (Mc)	-18	-18	18	18



Outplane Bending (Mc)	-116	116	116	-116
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	627	627	627	627

Outplane Stress Summary	466	737	737	541
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-5	-5	-5	-5
Inplane Bending (P)	-61	61	-61	61
Inplane Membrane (Mc)	-22	-22	22	22
Inplane Bending (Mc)	-383	383	383	-383
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1255	1340	1255	1340

Inplane Stress Summary	782	1758	1594	1035
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	17	17	17	17

Shear Stress Summary	11	11	23	23
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	782	1758	1594	1036
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-30	-30	30	30
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	117	220	117	220

Hoop Stress Summary	80	182	141	244
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-5	5	-5	5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-15	15	15	-15
Axial Pressure Stress	58	58	58	58

Axial Stress Summary	17	61	86	66
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	8	8	4	4

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	81	183	142	244

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	117	220	117	220
Hoop Stress Summary	92	194	129	232

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-5	5	-5	5
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-40	40	40	-40
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	58	58	58	58
Axial Stress Summary	-2	90	106	36

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	8	8

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	95	194	132	232

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	1255	1340	1255	1340	1255	1340	1255	1340
Circ. Pl (SUS)	-36	-36	24	24	-27	-27	17	17
Circ. Q (SUS)	-71	71	35	-35	-444	444	322	-322
Long. Pm (SUS)	627	627	627	627	627	627	627	627
Long. Pl (SUS)	-25	-25	15	15	-24	-24	12	12
Long. Q (SUS)	-244	244	122	-122	-134	134	98	-98
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	17	17	17	17	17	17	17	17
Pm (SUS)	1255	1340	1255	1340	1255	1340	1255	1340
Pm+Pl (SUS)	1219	1304	1279	1364	1228	1313	1272	1357
Pm+Pl+Q (Total)	1148	1375	1314	1329	784	1757	1594	1035

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1340.00	1407.13	Passed
Pm+Pl (SUS)	1364.03	2110.69	Passed
Pm+Pl+Q (TOTAL)	1757.14	4221.39	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	117	220	117	220	117	220	117	220
Circ. Pl (SUS)	-36	-36	24	24	-24	-24	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	58	58	58	58	58	58	58	58
Long. Pl (SUS)	-18	-18	18	18	-13	-13	13	13
Long. Q (SUS)	-20	20	10	-10	-45	45	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	117.0	220.0	117.0	220.0	117.0	220.0	117.0	220.0
Pm+Pl (SUS)	81.1	184.0	141.1	244.0	93.1	196.0	129.1	232.0
Pm+Pl+Q (Total)	82.0	184.5	141.3	244.1	93.3	196.2	131.5	232.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	220.00	1203.20	Passed
Pm+Pl (SUS)	244.02	1804.80	Passed
Pm+Pl+Q (TOTAL)	244.09	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	85.50 kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	85.50 kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	1.05 kgf/cm ²
Shell Thickness	ts	38.0000 mm
Shell Internal Corrosion Allowance	cas	3.1750 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1050.000 mm
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1407.21 kgf/cm ²
Shell Allowable Stress at Ambient		1407.21 kgf/cm ²

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	65.76 kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	65.76 kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 kgf/cm ²
Channel Thickness	tc	30.0000 mm
Channel Internal Corrosion Allowance	cac	3.1750 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1050.000 mm
Channel Design Temperature	TEMPC	160.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.21 kgf/cm ²
Channel Allowable Stress at Ambient		1407.21 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1418
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	4877.00 mm
Straight Tube Length (bet. inner tubsht faces) L		4662.00 mm
Design Temperature of the Tubes		160.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		942.17 kgf/cm ²
Tube Allowable Stress At Ambient		942.17 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1602.92 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	23.8120 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	508.176 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d



Tubesheet Design Metal Temperature T 160.00 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 215.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 3.1250 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.1250 mm
 Tubesheet Outside Diameter A 1500.000 mm

Dimension G for the Channel Side Gc 1091.500 mm
 Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Both
 Flange Outside Diameter A 1500.000 mm
 Flange Inside Diameter B 1050.000 mm
 Flange Face Outside Diameter Fod 1112.000 mm
 Flange Face Inside Diameter Fid 1050.000 mm
 Gasket Outside Diameter Go 1101.000 mm
 Gasket Inside Diameter Gi 1082.000 mm
 Small end Hub thk. g0 38.0000 mm
 Large end Hub thk. g1 58.0000 mm
 Gasket Factor, m 3.00
 Gasket Design Seating Stress y 703.07 kgf/cm²
 Flange Facing Sketch Code Sketch 1a
 Column for Gasket Seating Code Column II
 Gasket Thickness tg 3.1750 mm
 Full face Gasket Flange Option Program Selects

Bolting Information:

Diameter of Bolt Circle C 1350.000 mm
 Nominal Bolt Diameter dB 80.0000 mm
 Thread Series TEMA Metric Thread
 Number of Bolts n 24
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 1621.26 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1621.26 kgf/cm²
 Weld between Flange and Shell/Channel 0.0000 mm

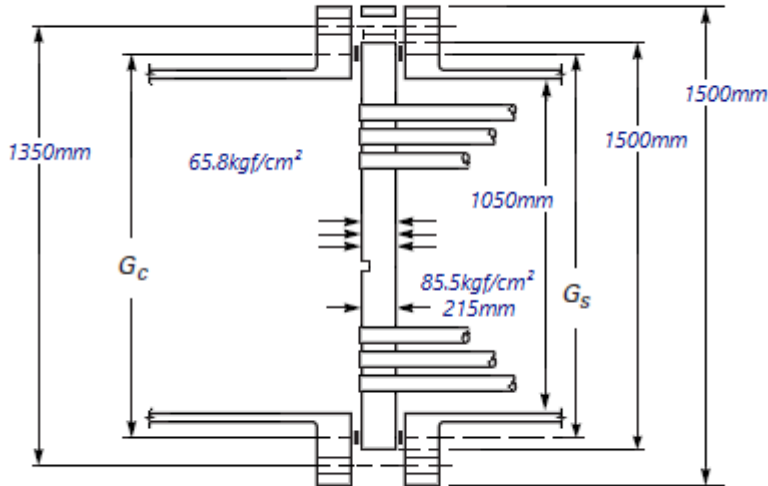
Alternate Flange Operating Bolt Load, Wm1 884613.88 kgf
 Alternate Flange Seating Bolt Load, Wm2 114515.77 kgf
 Alternate Flange Design Bolt Load, W 1586305.88 kgf

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with None
 Tubesheet Extended as Flange No



Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2023

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	160.0 °C	0.19841E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	160.0 °C	0.19841E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubesht. Design Temp.	160.0 °C	0.19841E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	160.0 °C	0.19841E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (65.8 \cdot (19/2 - 0)) / (942 \cdot 1 + 0.4 \cdot 65.8)$
 $= 0.6468 + 0.0000 = 0.6468 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 160.00 °C
 Elastic Modulus for Material 2024859.62 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	5170.18	9.03	50.0000	0.0134948	971.65
EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 152 \text{ kgf/cm}^2$						

Results for Req'd Thickness for Ext. Pressure (Tca):



TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.2812	19.05	5170.18	14.87	50.0000	0.0049753	953.56

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 954) / (3 \cdot 14.9) = 85.5 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.2812 mm
Allowable Internal Pressure at Corroded thickness	229.00 kgf/cm ²
Required Internal Design Pressure	65.76 kgf/cm ²
Allowable External Pressure at Corroded thickness	152.28 kgf/cm ²
Required External Design Pressure	85.50 kgf/cm ²
Required Thickness due to Shell Side pressure	1.2812 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2023

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	9.500 mm
Basic Gasket Width,	$b_0 = N / 2$	4.750 mm
Effective Gasket Width,	$b = b_0$	4.750 mm
Gasket Reaction Diameter,	$G = (G_o + G_i) / 2$	1091.500 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2023, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Compute the equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$
$$= 2 \cdot 508 + 19$$
$$= 1035.402 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$
$$= (23.8 - 19) / 23.8$$
$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Compute the ratio [ρ_s]:

$$= G_s / Do \text{ (Configurations d, e, f)}$$
$$= 1092 / 1035$$
$$= 1.0542$$

Compute the ratio [ρ_c]:

$$= G_c / Do \text{ (Configurations d)}$$
$$= 1092 / 1035$$
$$= 1.0542$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Compute the ratio [ρ]:

$$= l_{tx} / h$$
$$= 0 / 215$$
$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Compute the effective tube hole Diameter [d^*]:

$$= \max(dt - 2t_t (E t / E) \cdot (S t / S) \cdot (\rho); dt - 2t_t)$$
$$= \max(19 - 2 \cdot 2.11 (1984070 / 1984070) \cdot (942 / 1407) \cdot (0); 19 - 2 \cdot 2.11)$$
$$= 19.0500 \text{ mm}$$

Compute the effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot Do \cdot \rho) / (\pi \cdot Do^2)) }$$
$$= 23.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1035 \cdot 23.8) / (\pi \cdot 1035^2)) }$$



= 23.8120 mm

Compute the effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (23.8 - 19) / 23.8 \\ &= 0.2000 \end{aligned}$$

$$\begin{aligned} h/p &= 9.029061 ; \mu^* = 0.199983 ; E^* = 308184 \text{ kgf/cm}^2 \\ E^*/E &= 0.155329 ; \nu^* = 0.458596 \end{aligned}$$

Note: As h/p (9.029) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 1500/1035 \\ &= 1.4487 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (E \cdot \ln(K)) \\ &= (1 - 0.46) / 308184 \cdot (1984070 \cdot \ln(1.45)) \\ &= 1.2920 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$\begin{aligned} &= D_o^2/16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t] \\ &= 1035^2/16 \cdot [(1.05 - 1) \cdot (1.05^2 + 1) \cdot 85.5 - \\ &\quad (1.05 - 1) \cdot (1.05^2 + 1) \cdot 65.8] \\ &= 151221.8750 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M^*]

$$\begin{aligned} &= M_{ts} + W^* \cdot (G_c - G_s) / (2 \cdot \pi \cdot D_o) \\ &= 151222 + 1586305 \cdot (1092 - 1092) / (2 \cdot \pi \cdot 1035) \\ &= 151221.8750 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [M_p]:

$$\begin{aligned} &= (M^* - D_o^2/32 \cdot F(P_s - P_t)) / (1 + F) \\ &= (151222 - 1035^2/32 \cdot 1.29(85.5 - 65.8)) / (1 + 1.29) \\ &= -306623.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [M_o]:

$$\begin{aligned} &= M_p + D_o^2/64 \cdot (3 + \nu^*) \cdot (P_s - P_t) \\ &= -306624 + 1035^2/64 \cdot (3 + 0.46) \cdot (85.5 - 65.8) \\ &= 836431.8125 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|M_p|; |M_o|) \\ &= \max(|-306624|; |836432|) \\ &= 836431.8125 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h_g')^2) \\ &= 6 \cdot 836432 / (0.2 (215 - 6)^2) \\ &= 574.5074 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 1407 \\ &= 2814.26 \text{ kgf/cm}^2 \end{aligned}$$



Tube sheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 836432 / (0.2 \cdot (101 - 6)^2) \\ &= 2801.3965 \text{ kgf/cm}^2 \end{aligned}$$

Required tube sheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 101 + 0 + 0 \\ &= 100.6470 \text{ mm} \end{aligned}$$

Required tube sheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(101; 22.7) \\ &= 100.6470 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |85.5 - 65.8| \\ &= 19.730 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.2 \cdot 215 / 1035 \cdot \min(0.8 \cdot 1407; 0.533 \cdot 2221) \\ &= 186.985 \text{ kgf/cm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.2) \cdot (1035 / 1126) \cdot |85.5 - 65.8| + 0 \\ &= 22.6862 \text{ mm} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19 - 2.11) 2.11 \\ &= 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S_a \\ &= \pi \cdot 2.11 (19 - 2.11) 942 \\ &= 1057.971 \text{ kgf} \end{aligned}$$

Groove Weld Strength [F_g]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 942 \\ &= 1057.9706 \text{ kgf} \end{aligned}$$

Weld Strength Factor [f_w]:

$$\begin{aligned} &= S_a / S_w \text{ where } S_w = \min(S_a \text{ and } S_t) \\ &= 942 / 942 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [F_f]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 942 \\ &= 702.9219 \text{ kgf} \end{aligned}$$

Minimum Length of Weld [a_r]:

$$\begin{aligned} &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right) \\ &= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) 1 \cdot 1)} - 0.75 \cdot 19 \right) \end{aligned}$$



$$0.75 \cdot 19)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm
 Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1058 is >= tube strength 1058 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	574.5 <=	2814.3 kgf/cm ²	Ok
Tubesheet shear stress	118.8 <=	1125.7 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	100.647	215.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2023

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	178.440	15.803	1915 2814 396 1126	Ps+Pt D1	Ok
D2uc	202.581	20.544	2489 2814 515 1126	Ps+Pt D2	Ok
D3uc	100.647	4.741	575 2814 119 1126	Ps+Pt D3	Ok
D1c	181.536	15.803	1974 2814 408 1126	Ps+Pt-c D1	Ok
D2c	205.697	20.544	2566 2814 530 1126	Ps+Pt-c D2	Ok
D3c	103.604	4.741	592 2814 122 1126	Ps+Pt-c D3	Ok
Max:	205.6972	20.544 mm	0.912	0.471 (Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	205.697	215.000	Ok
Tube thickness :	1.281	2.110	Ok



Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	93.779	1.000	93.779	1.000
Tubesheet Shear Stress	181.548	1.000	181.548	1.000
Tube Pressure Stress	229.002	1.000	152.275	1.000
Minimum MAWP	93.779		93.779	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	96.648	1.000	96.648	1.000
Tubesheet Shear Stress	186.984	1.000	186.984	1.000
Tube Pressure Stress	229.002	1.000	153.156	1.000
Minimum MAPnc	96.648		96.648	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):
 = tubesheet thickness/4
 = 215/4
 = 53.750 mm

Note: This Element/Detail was specified as being Impact Tested.

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(65.8/96.6, 85.5/96.6)
 = 0.885

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: PWHT was required for at least 1 Nozzle/Pad in this vessel!**

Please review all reports carefully!

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